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1 Smooth Manifold

1.1 Topological Manifolds

Definition 1.1.1

Suppose M is a topological space. We say that M is a ***topological manifold of dimension n*** or a ***topological n-manifold*** if it has the following properties:

- M is a ***Hausdorff space***(T_2) : for every pair of distinct points $p, q \in M$, there are disjoint open subsets $U, V \subseteq M$ such that $p \in U$ and $q \in V$.
- M is ***second-countable***: there exists a countable basis for the topology of M.
- M is ***locally Euclidean of dimension n***: each point of M has a neighborhood that is homeomorphic to an open subset of $\mathbb{R}^n$.

Remark.
The third property means, more specifically, that for each $p \in M$ we can find

- an open subset $U \subseteq M$ containing p,
- an open subset $\hat{U} \subseteq \mathbb{R}^n$, and
- a homeomorphism $\varphi : U \to \hat{U}$.

Remark.
A topological manifold M is locally compact, locally path-connected, seperable, Lindëlof, σ-finite, T_3, T_4, and paracompact.

1.1.1 Coordinate Chart

Definition 1.1.2 ▶ Coordinate Chart

Let M be a topological n-***manifold***. A ***coordinate chart*** (or just a ***chart***) on M is a pair (U, φ), where U is an open subset of M and $\varphi : U \to \hat{U}$ is a homeomorphism from U to an open subset $\hat{U} = \varphi(U) \subseteq \mathbb{R}^n$.

Remark.

- By definition of a topological manifold, each point $p \in M$ is contained in the domain of some chart (U, φ).
- If $\varphi(p) = 0$, we say that the chart is ***centered at p***.
- If (U, φ) is any chart whose domain contains p, it is easy to obtain a new chart centered at p by subtracting the constant vector $\varphi(p)$.

1.2 Smooth Structures

Definition 1.2.1 ▶ Smoothly Compatible

Let M be a topological n-manifold.

- If $(U, \varphi), (V, \psi)$ are two charts such that $U \cap V \neq \varnothing$, the composite map $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is called the ***transition map from*** φ ***to*** ψ.
- Two charts (U, φ) and (V, ψ) are said to be ***smoothly compatible*** if either $U \cap V = \varnothing$ or the transition map $\psi \circ \varphi^{-1}$ is a diffeomorphism.

Remark.

- $\psi \circ \varphi^{-1}$ is a composition of homeomorphisms, and is therefore itself a homeomorphism.
- Since $\varphi(U \cap V)$ and $\psi(U \cap V)$ are open subsets of $\mathbb{R}^n$, smoothness of this map is to be interpreted in the ordinary sense of having continuous partial derivatives of all orders.

Definition 1.2.2 ▶ Atlas

We define an ***atlas for M*** to be a collection of charts whose domains cover M. An atlas $\mathcal{A}$ is called a ***smooth atlas*** if any two charts in $\mathcal{A}$ are smoothly compatible with each other.

Definition 1.2.3 ▶ Maximal Atlas

A smooth atlas $\mathcal{A}$ on M is ***maximal*** if it is not properly contained in any larger smooth atlas. This just means that any chart that is smoothly compatible with every chart in $\mathcal{A}$ is already in $\mathcal{A}$.

Definition 1.2.4 ▶ Smooth Structures and Smooth Manifolds

- If M is a topological manifold, a ***smooth structure on M*** is a maximal smooth atlas.
- A ***smooth manifold*** is a pair $(M, \mathcal{A})$, where M is a topological manifold and $\mathcal{A}$ is a smooth structure on M.

Lemma 1.2.5 ▶ Smooth Manifold Chart Lemma

Let M be a set, and suppose we are given a collection $\{U_\alpha\}$ of subsets of M together with maps $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$, such that the following properties are satisfied:

1. For each α, φ_α is a bijection between U_α and an open subset $\varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$.
2. For each α and β, the sets $\varphi_\alpha(U_\alpha \cap U_\beta)$ and $\varphi_\beta(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^n$.
3. Whenever $U_\alpha \cap U_\beta \neq \varnothing$, the map $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is smooth.
4. Countably many of the sets U_α cover M.
5. Whenever p, q are distinct points in M, either there exists some U_α containing both p and q or there exist disjoint sets U_α, U_β with $p \in U_\alpha$ and $q \in U_\beta$.

Then M has a unique smooth manifold structure such that each $(U_\alpha, \varphi_\alpha)$ is a smooth chart.

1.3 Manifolds with Boundary

Definition 1.3.1 ▶ Manifolds with Boundary

An ***n-dimensional topological manifold with boundary*** is a second-countable Hausdorff space M in which every point has a neighborhood homeomorphic either to an open subset of $\mathbb{R}^n$ or to a (relatively) open subset of $\mathbb{H}^n$.

Definition 1.3.2 ▶ Chart for Manifolds with Boundary

Let M be an n-dimensional topological manifold with boundary. An open subset $U \subseteq M$ together with a map $\varphi : U \to \mathbb{R}^n$ that is a homeomorphism onto an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$ will be called a ***chart for*** M.

- When it is necessary to make the distinction, we will call (U, φ) an ***interior chart*** if $\varphi(U)$ is an open subset of $\mathbb{R}^n$ (which includes the case of an open subset of $\mathbb{H}^n$ that does not intersect $\partial\mathbb{H}^n$).
- We call (U, φ) a ***boundary chart*** if $\varphi(U)$ is an open subset of $\mathbb{H}^n$ such that $\varphi(U) \cap \partial\mathbb{H}^n \neq \varnothing$.
- A boundary chart whose image is a set of the form $B_r(x) \cap \mathbb{H}^n$ for some $x \in \partial\mathbb{H}^n$ and $r > 0$ is called a ***coordinate half-ball***.

2 Smooth Maps

2.1 Smooth Functions and Smooth Maps

2.1.1 Smooth Maps Between Manifolds

Definition 2.1.1 ▶ Smooth Maps Between Manifolds

Let M, N be smooth manifolds, and let $F : M \to N$ be any map. We say that F is a ***smooth map*** if for every $p \in M$, there exist smooth charts (U, φ) containing p and (V, ψ) containing $F(p)$ such that $F(U) \subseteq V$ and the composite map $\psi \circ F \circ \varphi^{-1}$ is smooth from $\varphi(U)$ to $\psi(V)$.

Remark.

- By taking $N = V = \mathbb{R}^k$ and $\psi = \mathrm{Id} : \mathbb{R}^k \to \mathbb{R}^k$, we get the definition of smoothness of real-valued or vector-valed functions.

2.2 Partitions of Unity

Definition 2.2.1 ▶ Support

If f is any real-valued or vector-valued function on a topological space M, the ***support of f***, denoted by $\operatorname{supp} f$, is the closure of the set of points where f is nonzero:

$$\operatorname{supp} f = \overline{\{p \in M : f(p) \neq 0\}}.$$

- If $\operatorname{supp} f$ is contained in some set $U \subseteq M$, we say that f is ***supported in U***.
- A function f is said to be ***compactly supported*** if $\operatorname{supp} f$ is a compact set, denoted by $f \in C_0^\infty(M)$.

Remark.
Clearly, every function on a compact space is compactly supported.

Lemma 2.2.2

The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \leq 0, \end{cases}$$

is smooth.

Lemma 2.2.3

Given any real numbers r_1 and r_2 such that $r_1 < r_2$, there exists a smooth function $h : \mathbb{R} \to \mathbb{R}$ such that $h(t) \equiv 1$ for $t \leq r_1$, $0 < h(t) < 1$ for $r_1 < t < r_2$, and $h(t) \equiv 0$ for $t \geq r_2$.

Proof. Let f be the function of the previous lemma, and set

$$h(t) = \frac{f(r_2 - t)}{f(r_2 - t) + f(t - r_1)}.$$

Note that the denominator is positive for all t, because at least one of the expressions $r_2 - t$ and $t - r_1$ is always positive. The desired properties of h follow easily from those of f. $\square$

Lemma 2.2.4 ▶ Bump Function on $\mathbb{R}^n$

Given any positive real numbers $r_1 < r_2$, there is a smooth function $H : \mathbb{R}^n \to \mathbb{R}$ such that $H \equiv 1$ on $\bar{B}_{r_1}(0)$, $0 < H(x) < 1$ for all $x \in B_{r_2}(0) \setminus \bar{B}_{r_1}(0)$, and $H \equiv 0$ on $\mathbb{R}^n \setminus B_{r_2}(0)$.

Proof. Just set $H(x) = h(|x|)$, where h is the function of the preceding lemma. Clearly, H is smooth on $\mathbb{R}^n \setminus \{0\}$, because it is a composition of smooth functions there. Since it is identically equal to 1 on $B_{r_1}(0)$, it is smooth there too. $\square$

Theorem 2.2.5 ▶ Existence of Bump Function on Manifold

Let M be a smooth manifold, $K \subseteq M$ a compact set. $U \subseteq M$ is a open set such that $K \subseteq U$. Then there exists $\varphi \in C_0^\infty(M)$, such that $\operatorname{supp} \varphi \subseteq U$, $0 \leq \varphi \leq 1$ and $\varphi \equiv 1$ on K.

Proof. For each $x \in K$, take a chart (φ_x, U_x, V_x) centering x. We may assume that $V_x \subseteq B_3(0)$ and $U_x \subseteq U$ by contracting when necessary . Let $\tilde{U}_x := \varphi_x^{-1}(B_1(0))$. We define

$$f_x(p) := \begin{cases} H \circ \varphi_x(p), & p \in U_x \\ 0, & p \notin U_x \end{cases}$$

Where H is defined in Lemma 2.2.4 and we take $r_1 = 1, r_2 = 2$. It is straightforward to verify that $f_x \equiv 1$ on $\tilde{U}_x$ and $f_x \equiv 0$ on U_x^c, $\operatorname{supp} f_x = \varphi_x^{-1}\left(\overline{B_2(0)}\right) \subseteq U_x$. Since the homeomorphism keeps the closure $\varphi_x^{-1}\left(\overline{B_2(0)}\right) = \overline{\varphi_x^{-1}(B_2(0))}$.

Now K has a open cover $\{\tilde{U}_x\}$. Since K is compact, there are finitely many $\tilde{U}_{x_i}$ such that

$$\bigcup_{i=1}^{N} \tilde{U}_{x_i} \supseteq K$$

We define

$$\psi := \sum_{i=1}^{N} f_{x_i}$$

Then $\psi \geq 1$ on K, $\operatorname{supp} \psi \subseteq U$.

Let $\rho : \mathbb{R} \to [0, 1]$ be a smooth function such that

$$\rho(t) = \begin{cases} 0, & t \leq 1/2 \\ 1, & t \geq 1 \end{cases}$$

and ρ is monotonically increasing on $(1/2, 1)$. Now, we define our final function $\varphi : M \to \mathbb{R}$ as the composition

$$\varphi := \rho \circ \psi$$

$\square$

Definition 2.2.6 ▸ Partitions of Unity

Suppose M is a topological space, and let $\mathcal{X} = (X_\alpha)_{\alpha \in A}$ be an arbitrary open cover of M, indexed by a set A. A ***partition of unity subordinate to*** $\mathcal{X}$ is an indexed family $(\psi_\alpha)_{\alpha \in A}$ of continuous functions $\psi_\alpha : M \to \mathbb{R}$ with the following properties:

1. $0 \le \psi_\alpha(x) \le 1$ for all $\alpha \in A$ and all $x \in M$.
2. $\operatorname{supp} \psi_\alpha \subseteq X_\alpha$ for each $\alpha \in A$.
3. The family of supports $(\operatorname{supp} \psi_\alpha)_{\alpha \in A}$ is locally finite, meaning that every point has a neighborhood that intersects $\operatorname{supp} \psi_\alpha$ for only finitely many values of α.
4. $\sum_{\alpha \in A} \psi_\alpha(x) = 1$ for all $x \in M$.

If M is a smooth manifold with or without boundary, a ***smooth partition of unity*** is one for which each of the functions ψ_α is smooth.

Remark.

- Because of the local finiteness condition (3), the sum in (4) actually has only finitely many nonzero terms in a neighborhood of each point, so there is no issue of convergence.
- For a manifold M, M is Lindelöf Space, we can always assmue that $(X_\alpha)_{\alpha \in A}$ is a countable colletction, thus the $(\psi_\alpha)_{\alpha \in A}$ is as well .

Theorem 2.2.7 ▸ Existence of Partitions of Unity

Suppose M is a smooth manifold with or without boundary, and $\mathcal{X} = (X_\alpha)_{\alpha \in A}$ is any indexed open cover of M. Then there exists a smooth partition of unity subordinate to $\mathcal{X}$.

Applications

Proposition 2.2.8 ▸ Bump function Support on a Closed Set

Let M be a manifold, $A \subseteq M$ be a closed set, $U \supseteq A$ is an open set. Then there exists $\varphi \in C_0^\infty(M)$ such that $0 \le \varphi \le 1$, and $\varphi \equiv 1$ on A .

Proof. $\{U, M \setminus A\}$ is a open cover of M. Then there exists a partition of unity $\{\rho_1, \rho_2\}$ subordinate to it. Then ρ_1 is just the function we need. □

Theorem 2.2.9 ▸ Stone-Weierstrass

Let X be a compact Hausdorff Space. $\mathcal{A} \subseteq C(X, \mathbb{R})$ is a subalgebra . Then $\mathcal{A}$ is dense in $C(X, \mathbb{R})$, if and only if it satisfies

1. Nowhere vanishing: For each $x \in X$, there exists $g \in \mathcal{A}$ such that $g(x) \neq 0$.
2. Separates points: For distinct $x, y \in X$, there exists $f \in \mathcal{A}$ such that

$f(x) \neq g(x)$.

Theorem 2.2.10 ▶ Whitney

Let M be a smooth manifold. For each continuous function $g \in : M \to \mathbb{R}, \delta : M \to \mathbb{R}_+$, there exists a smooth function $f : M \to \mathbb{R}$, such that

$$|f(p) - g(p)| < \delta(p), \quad \forall p \in M$$

Theorem 2.2.11 ▶ Whitney Extension

Let M be a smooth manifold. $A \subseteq M$ is a closed set. For continuous $g : M \to \mathbb{R}, \delta : M \to \mathbb{R}_+$, where g is smooth on A. Then there exists smooth function $f : M \to \mathbb{R}$, such that

1. $$f(p) = g(p), \quad \forall p \in A$$
2. $$|f(p) - g(p)| < \delta(p), \quad \forall p \in M$$

3

Tangent Vectors

3.1 Tangent Space

3.1.1 Geometric Tangent Vectors

Definition 3.1.1

If a is a point of $\mathbb{R}^n$, a map $w : C^\infty(\mathbb{R}^n) \to \mathbb{R}$ is called a ***derivation at a*** if it is linear over $\mathbb{R}$ and satisfies the following product rule:

$$w(fg) = f(a)wg + g(a)wf.$$

Let $T_a\mathbb{R}^n$ denote the set of all derivations of $C^\infty(\mathbb{R}^n)$ at a. Clearly, $T_a\mathbb{R}^n$ is a vector space under the operations

$$(w_1 + w_2)f = w_1 f + w_2 f, \quad (cw)f = c(wf).$$

Proposition 3.1.2

Let $a \in \mathbb{R}^n$.

(a) For each geometric tangent vector $v_a \in \mathbb{R}^n_a$, the map $D_v|_a : C^\infty(\mathbb{R}^n) \to \mathbb{R}$ defined by

$$D_v|_a f = D_v f(a) = \frac{d}{dt}\bigg|_{t=0} f(a + tv).$$

is a derivation at a.

(b) The map $v_a \mapsto D_v|_a$ is an isomorphism from $\mathbb{R}^n_a$ onto $T_a\mathbb{R}^n$.

3.1.2 Tangent Vectors on Manifolds

Definition 3.1.3

Let M be a smooth manifold with or without boundary, and let p be a point of M. A linear map $v : C^\infty(M) \to \mathbb{R}$ is called a ***derivation at p*** if it satisfies

$$v(fg) = f(p)vg + g(p)vf \quad \text{for all } f, g \in C^\infty(M). \tag{3.1}$$

The set of all derivations of $C^\infty(M)$ at p, denoted by T_pM, is a vector space called the ***tangent space to M at p***. An element of T_pM is called a ***tangent vector at p***.

Lemma 3.1.4 ▶ Properties of Tangent Vectors on Manifolds

Suppose M is a smooth manifold with or without boundary, $p \in M$, $v \in T_pM$, and $f, g \in C^\infty(M)$.

(a) If f is a constant function, then $vf = 0$.
(b) If $f(p) = g(p) = 0$, then $v(fg) = 0$.

3.2 The Differential of a Smooth Map

3.2.1 Differential on Euclidean Space

Lemma 3.2.1

Let $U \subseteq \mathbb{R}^m, V \subseteq \mathbb{R}^n$ are open sets. $f : U \to V$ is smooth. Take $a \in v$, define a map

$$\begin{aligned} (df)_a : T_aU &\to T_{f(a)}V \\ (df_a)(D_v^a) &\mapsto D_w^{f(a)}, \quad w = J_f(a)v \end{aligned}$$

Then for each $g \in C^\infty(V)$, we have

$$((df_a)(v))(g) = D_v^a(g \circ f)$$

Note.
$(df_a)(v)$ 对应的导子恰好是在点 $f(a)$ 处将 $g \in C^\infty(V)$ 映到 $D_v^a(g \circ f)$ 的导子.

Definition 3.2.2

If M and N are smooth manifolds with or without boundary and F :

$M \to N$ is a smooth map, for each $p \in M$ we define a map

$$dF_p : T_pM \to T_{F(p)}N,$$

called the ***differential of F at p***. Given $v \in T_pM$, we let $dF_p(v)$ be the derivation at $F(p)$ that acts on $f \in C^\infty(N)$ by the rule

$$dF_p(v)(f) = v(f \circ F).$$

Proposition 3.2.3 ▸ Properties of Differentials

Let M, N, and P be smooth manifolds with or without boundary, let $F : M \to N$ and $G : N \to P$ be smooth maps, and let $p \in M$.

(a) $dF_p : T_pM \to T_{F(p)}N$ is linear.
(b) $d(G \circ F)_p = dG_{F(p)} \circ dF_p : T_pM \to T_{G\circ F(p)}P$.
(c) $d(\mathrm{Id}_M)_p = \mathrm{Id}_{T_pM} : T_pM \to T_pM$.
(d) If F is a ***diffeomorphism***, then $dF_p : T_pM \to T_{F(p)}N$ is an isomorphism, and $(dF_p)^{-1} = d(F^{-1})_{F(p)}$.

3.3 Tangent Bundle

Definition 3.3.1

Given a smooth manifold M with or without boundary, we define the ***tangent bundle of*** M, denoted by TM, to be the disjoint union of the tangent spaces at all points of M:

$$TM = \bigsqcup_{p \in M} T_pM.$$

Remark.

- ***projection map*** $\pi : TM \to M$, $\pi(p, v) = p$.

Proposition 3.3.2

For any smooth n-manifold M, then tangent bundle TM has a natural topology and smooth structure that make it into a $2n$-dimensional smooth manifold. With respect to this structure, the projection $\pi : TM \to M$ is smooth.

Proof. Given a chart (U, φ) for M, $\pi^{-1}(U) \subseteq TM$ is the set of all tangent vectors to M at all points of U. Define a map $\tilde{\varphi} : \pi^{-1}(U) \to \mathbb{R}^{2n}$ by

$$\tilde{\varphi}\left(v^i \left.\frac{\partial}{\partial x^i}\right|_p\right) = \left(x^1(p), \cdots, x^n(p), v^1, \cdots, v^n\right)$$

$\square$

4 Submersions, Immersions and Embedings

4.0.1 Local Diffeomorphism

Definition 4.0.1

If M and N are smooth manifolds with or without boundary, a map $F : M \to N$ is called a ***local diffeomorphism*** if every point $p \in M$ has a neighborhood U such that $F(U)$ is open in N and $F|_U : U \to F(U)$ is a diffeomorphism.

Theorem 4.0.2 ▶ Inverse Function Theorem for Manifolds

Suppose M and N are smooth manifolds, and $F : M \to N$ is a smooth map. If $p \in M$ is a point such that dF_p is invertible, then there are connected neighborhoods U_0 of p and V_0 of $F(p)$ such that $F|_{U_0} : U_0 \to V_0$ is a diffeomorphism.

Note.
也就是说, 光滑映射在微分映射可逆的点附近是微分同胚.

4.1 Submersions

Definition 4.1.1

- If $\pi : M \to N$ is any continuous map, a ***section of*** π is a continuous right inverse for π, i.e., a continuous map $\sigma : N \to M$ such that $\pi \circ \sigma = \mathrm{Id}_N$:

$$\begin{tikzcd} M \arrow[d, "\pi"', shift right] \\ N \arrow[u, bend right, "\sigma"'] \end{tikzcd}$$

- A ***local section of*** π is a continuous map $\sigma : U \to M$ defined on some open subset $U \subseteq N$ and satisfying the analogous relation $\pi \circ \sigma = \mathrm{Id}_U$.

Theorem 4.1.2 ▸ Local Section Theorem

Suppose M and N are smooth manifolds and $\pi : M \to N$ is a smooth map. Then π is a smooth submersion if and only if every point of M is in the image of a smooth local section of π.

Proof sketch.
光滑淹没的标准坐标表示 $\pi\left(x^1, \cdots, x^n, x^{n+1}, \cdots, x^m\right) = \left(x^1, \cdots, x^n\right)$ 下，将坐标邻域缩小成方块 $C_\varepsilon = \left\{x : \left|x^i\right| < \varepsilon, i = 1, \cdots, m\right\}$，那么坐标像也是方片 $C'_\varepsilon = \left\{y : \left|y^i\right| < \varepsilon, i = 1, \cdots, n\right\}$. 通过 C'_ε 上的点末尾坐标补零 $\sigma\left(x^1, \cdots, x^n\right) = \left(x^1, \cdots, x^n, 0, \cdots, 0\right)$, 可以方便地给出一个截面.
反过来, 若 p 点位于某个截面像, 则 $\pi \circ \sigma = \mathrm{Id}_U$ 求导 $d\pi_p \circ d\sigma_q = \mathrm{Id}_{T_qN}$ 立即给出 π 在 p 点是淹没.

Proposition 4.1.3 ▸ Properties of Smooth Submersions

Let M and N be smooth manifolds, and suppose $\pi : M \to N$ is a smooth submersion. Then π is an open map, and if it is surjective it is a quotient map.

Proof sketch.
截面性质 $\pi \circ \sigma = \mathrm{Id}_U$ 给出 $\pi \circ \sigma\left(\sigma^{-1}(W)\right) \subseteq \pi(W)$. 截面的连续性给出 $\sigma^{-1}(W)$ 是开集, 若 W 是开集. 由此给出像点的开邻域, 说明淹没是开的.
连续而开的映射是商映射.

Theorem 4.1.4 ▸ Characteristic Property of Surjective Smooth Submersions

Suppose M and N are smooth manifolds, and $\pi : M \to N$ is a surjective smooth submersion. For any smooth manifold P with or without boundary, a map $F : N \to P$ is smooth if and only if $F \circ \pi$ is smooth:

$$\begin{tikzcd} M \arrow[d, "\pi"'] \arrow[dr, "F\circ\pi"] & \\ N \arrow[r, "F"'] & P. \end{tikzcd}$$

Proof sketch.
局部截面 σ 的存在性, 允许我们局部地将 F 提升到 M 上 $F \circ \pi$, 使得它可以被 σ 还原: $F|_U = (F \circ \pi) \circ \sigma$.

Theorem 4.1.5 ▸ Passing Smoothly to the Quotient

Suppose M and N are smooth manifolds and $\pi : M \to N$ is a surjective smooth submersion. If P is a smooth manifold with or without boundary and $F : M \to P$ is a smooth map that is constant on the fibers of π, then there exists a unique smooth map $\tilde{F} : N \to P$ such that $\tilde{F} \circ \pi = F$:

$$\begin{tikzcd} M \arrow[d, "\pi"'] \arrow[dr, "F"] & \\ N \arrow[r, dashed, "\tilde{F}"'] & P. \end{tikzcd}$$

Proof sketch.
满的光滑淹没是商映射, 商的泛性质刻画出唯一的满足交换图的连续映射 $\tilde{F}$, 再通过满的光滑淹没刻画其连续性.

5 Submanifolds

5.1 Embedded Submanifolds

5.1.1 Slice Charts for Embedded Submanifolds

Definition 5.1.1

If U is an open subset of $\mathbb{R}^n$ and $k \in \{0, \ldots, n\}$, a ***k-dimensional slice of U*** (or simply a ***k-slice***) is any subset of the form

$$S = \{(x^1, \ldots, x^k, x^{k+1}, \ldots, x^n) \in U : x^{k+1} = c^{k+1}, \ldots, x^n = c^n\}$$

for some constants $c^{k+1}, \ldots, c^n$.

Definition 5.1.2 ▸ local k-slice condition

Let M be a smooth n-manifold.

- Let (U, φ) be a smooth chart on M. If S is a subset of U such that $\varphi(S)$ is a k-slice of $\varphi(U)$, then we say that S is a ***k-slice of U***.
- Given a subset $S \subseteq M$ and a nonnegative integer k, we say that S satisfies the ***local k-slice condition*** if each point of S is contained in the domain of a smooth chart (U, φ) for M such that $S \cap U$ is a single k-slice in U.
- Any such chart is called a ***slice chart for S in M***, and the corresponding coordinates $(x^1, \ldots, x^n)$ are called ***slice coordinates***.

5.1.2 Level Sets

Definition 5.1.3

If $\Phi : M \to N$ is a mooth map.

- A point $p \in M$ is said to be a ***regular point of $V\phi$*** if $d\Phi_p : T_pM \to T_{\Phi(p)}N$ is srjective; it is a ***critical point of Φ*** otherwise.
- A point $c \in N$ is said to be a ***regular value of Φ*** if every point of the level set $\Phi^{-1}(c)$ is a regular point, and a ***critical value*** otherwise.

Theorem 5.1.4 ▸ Constant-Rank Level Set Theorem

Let M and N be smooth manifolds, and let $\Phi : M \to N$ be a smooth map with constant rank r. Each level est of Φ is a properly embedded submanifold of codimension r in M.

Proof. Write $m = \dim M$, $n = \dim N$, and $k = m - r$. Let $c \in M$ be arbitrary, let S denote the level set $\Phi^{-1}(c) \subseteq M$. From the rank theorem, for each $p \in S$, there are smooth charts (U, φ) centered at p and (V, ψ) centered at $c = \Phi(p)$ in which Φ has a coordinate representation of the form

$$\psi \circ \Phi \circ \varphi^{-1}\left(x^1, \cdots, x^m\right) = \left(x^1, \cdots, x^r, 0, \cdots, 0\right)$$

and therefore $S \cap U$, then

$$\begin{aligned} q \in S \iff \Phi(q) = c &\iff \left(x^1(q), \cdots, x^r(q), 0, \cdots, 0\right) = (0, \cdots, 0) \\ &\iff x^1(q) = \cdots = x^r(q) = 0 \end{aligned}$$

Thus

$$S \cap U = \left\{\left(x^1, \cdots, x^r, x^{r+1}, \cdots, x^m\right) \in U : x^1 = \cdots = x^r = 0\right\}$$

is the k-slice. So S satisfies the local k-slice condition, so it is an embedded submanifold of dimension k. □

5.2 Restricting Maps to Submanifolds

Theorem 5.2.1 ▸ Restricting the Domain of a Smooth Map

If M and N are smooth manifolds with or without boundary, $F : M \to N$ is a smooth map, and $S \subseteq M$ is an immersed or embedded submanifold , then $F|_S : S \to N$ is smooth.

Theorem 5.2.2 ▸ Restricting the Codomain of a Smooth Map

Suppose M is a smooth manifold (without boundary), $S \subseteq M$ is an immersed submanifold, and $F : N \to M$ is a smooth map whose image is contained in S . If F is continuous as a map from N to S, then $F : N \to S$ is smooth.

5.3 The Tangent Space to a Submanifold

Definition 5.3.1

Let M be a smooth manifold with or without boundary, and let $S \subseteq M$ be an ***immersed*** or embedded submanifold. Since the inclusion map $\iota : S \hookrightarrow M$ is a smooth immersion, at each point $p \in S$ we have an injective linear map $d\iota_p : T_pS \to T_pM$. In terms of derivations, this injection works in the following way: for any vector $v \in T_pS$, the image vector $\tilde{v} = d\iota_p(v) \in T_pM$ acts on smooth functions on M by

$$\tilde{v} f = d\iota_p(v) f = v(f \circ \iota) = v\big(f|_S\big).$$

Remark.
We adopt the convention of ***identifying*** T_pS with its image under this map, thereby thinking of T_pS as a certain linear subspace of T_pM

Proposition 5.3.2

Suppose M is a smooth manifold with or without boundary, $S \subseteq M$ is an immersed or embedded submanifold, and $p \in S$. A vector $v \in T_pM$ is in T_pS if and only if there is a smooth curve $\gamma : J \to M$ whose image is contained in S, and which is also smooth as a map into S, such that $0 \in J$, $\gamma(0) = p$, and $\gamma'(0) = v$.

Proof. $\Longleftarrow$: If there is $\gamma : J \to M$ satisfying

1. $\operatorname{Im}(\gamma) \subseteq S$
2. $\tilde{\gamma} : J \to S$ is smooth, where $\iota \circ \tilde{\gamma} = \gamma$
3. $\gamma(0) = p, \gamma'(0) = v$.

Then since

$$v = \gamma'(0) = (\iota \circ \tilde{\gamma})'(0) = d\iota_p(\tilde{\gamma}'(0))$$

$\tilde{\gamma}$ is a smooth curve on S, $\tilde{\gamma}'(0) \in T_pS$, thus $v \in T_pS \subseteq T_pM$.

$\Longrightarrow$ if $v \in T_pS \subseteq T_pM$, then there exsits $u \in T_pS$ such that $d\iota_p(u) = v$. There exists a smooth curve $\tilde{\gamma} : J \to S$ such that $\tilde{\gamma}(0) = p$, $\tilde{\gamma}'(0) = u$. Let $\gamma = \iota \circ \tilde{\gamma}$, then

$$\operatorname{Im}\gamma \subseteq S, \quad \gamma(0) = \iota(\tilde{\gamma}(0)) = \iota(p) = p, \quad \gamma'(0) = \mathrm{d}\iota_p(\tilde{\gamma}'(0)) = d\iota_p(u) = v$$

$\square$

Proposition 5.3.3

uppose M is a smooth manifold, $S \subseteq M$ is an ***embedded*** submanifold, and $p \in S$. As a subspace of T_pM, the tangent space T_pS is characterized by

$$T_pS = \{v \in T_pM : vf = 0 \text{ whenever } f \in C^\infty(M) \text{ and } f|_S = 0\}.$$

Proof. First we assume that $v \in T_pS \subseteq T_pM$. This means, more precisely, there exists $w \in T_pS$ such that $d\iota_p(w) = v$. Then for each $f \in C^\infty(M)$ with $f|_S = 0$, we have

$$vf = d\iota_p(w) f = w(f \circ \iota) = 0$$

Conversely, if $v \in T_pM$ satisfies $vf = 0$ whenever $f \in C^\infty(M)$ and $f|_S = 0$. Let $(x^1, \cdots, x^n)$ be slice coordinates for S in some neighborhood U of p , so that $U \cap S$ is the subset of U where $x^{k+1} = \cdots = x^n = 0$, and $(x^1, \cdots, x^k)$ are coordinates for $S \cap U$. Then $\iota : S \cap U \hookrightarrow M$ is represented

$$\iota(x^1, \cdots, x^k) = (x^1, \cdots, x^k, 0, \cdots, 0)$$

Write

$$v = \sum_{i=1}^n v^i \left.\frac{\partial}{\partial x^i}\right|_p ,$$

Let $\varphi \in C^\infty(M)$ be a bump supported in U and is equal to 1 in a neighborhood of p . Take any $j > k$, let extend $f = \varphi(x) x^j$ to be zero on $M \setminus \operatorname{supp}\varphi$. Then f is identically zero on S, we have

$$0 = vf = \varphi(p) v^j = v^j$$

$\square$

Proposition 5.3.4

Suppose M is a smooth manifold and $S \subseteq M$ is an embedded submanifold. If $\Phi : U \to N$ is any local defining map for S, then $T_pS = \ker d\Phi_p : T_pM \to T_{\Phi(p)}N$ for each $p \in S \cap U$.

Proof. Let $\iota : S \hookrightarrow M$, then $\Phi \circ \iota$ is identically zero on $U \cap S$. Then $d\Phi_p \circ d\iota_p = 0$, thus $\operatorname{Im} d\iota_p \subseteq \ker d\Phi_p$. The other side, since $d\Phi_p$ is surjective, we have

$$\dim \ker d\Phi_p = \dim T_pM - \dim T_{\Phi(p)}N = \dim T_pS = \dim \operatorname{Im} d\iota_p$$

Thus $\operatorname{Im} d\iota_p = \ker d\Phi_p$. Since $\operatorname{Im} d\iota_p$ is identified with T_pS, we have $T_pS = \ker d\Phi_p$. $\square$

6 Sard's Theorem

6.1 Set of Measure Zero

Lemma 6.1.1

Suppose $A \subseteq \mathbb{R}^n$ is a compact subset whose intersection with $\{c\} \times \mathbb{R}^{n-1}$ has $(n-1)$- dimensional measure zero for every $c \in \mathbb{R}$. Then A has a n - dimensional measure zero .

Proof sketch.
把 A 的每一个“片” $A_c := \left\{(c,x) \in \{c\} \times \mathbb{R}^{n-1} : (c,x) \in A\right\}$ 的开覆盖, 缩短成里面的一个开的“方片”. 这些“方片”构成 A 的一个开覆盖, 取出一个有限子覆盖计算其 Volume, 将“方片”的相交部分抹去, 使得总体积可控.

Proof. Take $[a,b] \subseteq \mathbb{R}$ such that $A \subseteq [a,b] \times \mathbb{R}^{n-1}$.

Let $A_c := \left\{(c,x) \in \{c\} \times \mathbb{R}^{n-1} : (c,x) \in A\right\}$. The projection from A to $[a,b]\times R^{n-1}$ is continious map from a compact set to a Hausdorff space, thus is a proper map . It follows that A_c is a compact subspace of A for all $c \in [a,b]$.

Take any $\delta > 0$. There exists a finite open cover $C_1, \cdots, C_k$ of A_c , such that the total volumn is less than δ . Denote the open subset $C_1 \cup \cdots \cup C_k \subseteq A_c$ by U_c . We claim that there is a open interval J_c containing c and contained in U_c , such that A_c is contained in the intersection of A with $J_c \times \mathbb{R}^{n-1}$: otherwise, there is a suquence $\{(c_i, x_i)\}$, such that $c_i \to c$, $x_i \notin U_c$; but is passing to a subsequence $x_i \to A_c \setminus U_c$, which is a contradiction to $A_c \subseteq U_c$.

The intervals $\{J_c : c \in [a,b]\}$ forms a open cover of $[a,b]$, so there is a finitely many $c_1 < c_2 < \cdots < c_m$ such that $J_{c_1}, \cdots, J_{c_m}$ forms a finite open cover of A . We may assume that the total length is less than $2\,|b-a|$ by shrinking the intersection of the members. Then $J_{c_1}, \cdots, J_{c_m}$ is a open cover of A with total volumn less than $2\delta\,|b-a|$, which can be arbitrally small . Hence A has a n- dimensional measure zero . $\square$

Proposition 6.1.2

Suppose A is an open or closed subset of $\mathbb{R}^{n-1}$ or $\mathbb{H}^{n-1}$, and $f : A \to \mathbb{R}$ is a continuous function. Then the graph of f has measure zero in $\mathbb{R}^n$.

Proof sketch.
对于 A 是紧集的情况, 令 $P^n = \{\text{Sets are n-dimensional measure zero}\}$. 上面的引理给出了 $\forall c, P_c^{n-1} \implies P^n$ 的结论. 因此降维操作是无损的, 从而归纳法有效.

Proof. First assume that A is compact . We prove the theorem in this case by induction on n . When $n = 1$, it is trivial because the graph of f is at most a point. To prove the inductinve step, we appeal to lemma above. For each $c \in \mathbb{R}$, the intersection of the graph of f with $\{c\} \times \mathbb{R}^{n-1}$ is just the graph of the restriction of f to $\left\{x \in A : x^1 = c\right\}$, which is in turn the graph of continuous functions of $(n-2)$ variables. It follows by induction that each such graph has $(n-1)$-dimensional measure zero , and thus by lemma above , the graph of f itself has n -dimensional measure zero .

If A is noncompact , it is a countable union of compact subsets, so the graph of f is a countable union of sets of measure zero. □

Corollary 6.1.3

Every proper affine subspace of $\mathbb{R}^n$ has measure zero in $\mathbb{R}^n$.

Proposition 6.1.4 ▸ Diffeomorphism-invariant of Measure Zero

Supposer $A \subseteq \mathbb{R}^n$ has measure zero and $F : A \to \mathbb{R}^n$ is a smooth map . Then $F(A)$ has measure zero .

Definition 6.1.5

If M is a smooth n- manifold with or without boundary , we say that a subset $A \subseteq M$ has ***measure zero in*** M if for every smooth chat (U, φ) for M , the subset $\varphi(A \cap U) \subseteq \mathbb{R}^n$ has n- dimensional measure zero .

Remark.

Equivalently, $\forall \varepsilon > 0$, there are countably many charts (U_i, φ_i), such that

$$A \subseteq \bigcup_{i=1}^{\infty} U_i, \quad \sum_{i=1}\left[\left(\varphi_i^{-1}\right)_{\sharp}\mu\right](U_i) < \varepsilon$$

Lemma 6.1.6

Let M be a smooth n-manifold with or without boundary and $A \subseteq M$. Suppose that for some collection $\{(U_\alpha, \varphi_\alpha)\}$ of smooth charts whose domains cover A, $\varphi_\alpha(A \cap U_\alpha)$ has measure zero in $\mathbb{R}^n$ for each α . Then A has measure zero in M .

Proof. Let (V, ψ) be any smooth chat for M . We need to show that $\psi(A \cap V)$ has measure zero . There exists countably many U_α's covering $A \cap V$. We write

$$\psi(A \cap V \cap U_\alpha) = \left(\psi \circ \varphi_\alpha^{-1}\right) \circ \varphi_\alpha(A \cap V \cap U_\alpha)$$

$\varphi_\alpha(A \cap V \cap U_\alpha)$ is a subset of measure zero set $\varphi_\alpha(A \cap U_\alpha)$, thus has measure zero as well. It follows from Proposition 6.1.4 that $\psi(A \cap V \cap U_\alpha)$ has measure zero . Then $\psi(A \cap V)$ is a countable unions of measure zero sets , thus is measure zero as well. □

Exercise 6.1.7

Let M be a smooth manifold with or without boundary. Show that a countable union of sets of measure zero in M has measure zero.

Proof. Suppose $A_1, A_2, \cdots$ are sets of measure zero in M . Let $A = \bigcup_{i=1}^{\infty} A_i$. Take charts $\left(U_{ij}, \varphi_{ij}\right), j = 1, 2, \cdots$ such that $A_i \subseteq \bigcup_{j=1}^{\infty} U_{ij}$. Then $\left\{U_{ij}\right\}_{i,j}^{\infty}$ are countably many open sets that containing A , such that for each i, j ,

$$\varphi_{ij}\left(A \cap U_{ij}\right) \subseteq \bigcup_{i=1}^{\infty} \varphi_{ij}\left(A_i \cap U_{ij}\right)$$

is a countable union of measure zero sets, thus itself is measure zero. Then it follows from Lemma 6.1.6 that A is of measure zero. □

Proposition 6.1.8

Suppoes M is a smooth manifold with or without boundary and $A \subseteq M$ has measure zero in M. Then $M \setminus A$ is dense in M.

Proof. If $M \setminus A$ is not dense. Then A contains a nonempty open subset of M, which implies that there is a chat (V, ψ) , such that $\psi(A \cap V)$ is a nonemtpy open subsets of $\mathbb{R}^n$. It is imposible that $\psi(A \cap V)$ has measure zero at the same time that it is open. □

6.2 Sard's Theorem

Theorem 6.2.1 ▶ Sard's Theorem

Suppose M and N are smooth manifolds with or without boundary and $F : M \to N$ is a smooth map. Then the set of critical values of F has measure zero in N.

6.3 Tubular Neiborhoods

6.4 Transversality

Definition 6.4.1 ▶ Intersect Transversely

- Two embedded submanifolds $S, S' \subseteq M$ are said to ***intersect transversely*** if for each $p \in S \cap S'$, then tangent spaces T_pS and T_pS' togegher span T_pM .
- Let $F : N \to M$ be a smooth map and $S \subseteq M$ be an embedded submanifold. We say F is ***transverse*** to S, if for every $x \in F^{-1}(S)$, the spaces $T_{F(x)}S$ and $dF_x(T_xN)$ together span $T_{F(x)}M$.

Remark.
$S, S' \subseteq M$ are intersect transversely, if and only if the inclusion map $\iota : S \hookrightarrow M$ is transverse to S'.

Theorem 6.4.2

Suppose N and M are smooth manifolds and $S \subseteq M$ is an embedded submanifold.

(a) If $F : N \to M$ is a smooth map that is transverse to S, then $F^{-1}(S)$ is an embedded submanifold of N whose codimension is equal to the codimension of S in M.
(b) If $S' \subset M$ is an embedded submanifold that intersects S transversely, then $S \cap S'$ is an embedded submanifold of M whose codimension is equal to the sum of the codimensions of S and S'.

Corollary 6.4.3

Suppose N and M are smooth manifolds, $S \subseteq M$ is an embedded submanifold of codimension k, and $F : N \to M$ is a submersion. Then $F^{-1}(S)$ is an embedded codimension-k submanifold of N.

Proof. If F is a submersion , it is automatically transverse to every embedded submanifold $S \subseteq M$. $\square$

7 Smooth Vector Fields

7.1 Vector Fields on Manifold

Definition 7.1.1

If M is a smooth manifold with or without boundary, a ***vector field on*** M is a section of the map $\pi : TM \to M$. More concretely, a vector field is a continuous map $X : M \to TM$, usually written $p \mapsto X_p$, with the property that

$$\pi \circ X = \mathrm{Id}_M, \tag{7.1}$$

or equivalently, $X_p \in T_pM$ for each $p \in M$.

Remark.

- For a "vector field" that may be not contiuous, we call it ***rough vector field on*** M .
- Under the canonical smooth structure of M and TM , we can define the smoothness of X.

Definition 7.1.2 ▸ component functions of a vector field

Suppose M is a smooth n-manifold (with or without boundary). If $X : M \to TM$ is a rough vector field and $(U, (x^i))$ is any smooth coordinate chart for M, we can write the value of X at any point $p \in U$ in terms of the coordinate basis vectors:

$$X_p = X^i(p) \frac{\partial}{\partial x^i}\bigg|_p . \tag{7.2}$$

This defines n functions $X^i : U \to \mathbb{R}$, called the ***component functions of*** X in the given chart.

Proposition 7.1.3 ▶ Smoothness Criterion for Vector Fields

Let M be a smooth manifold with or without boundary, and let $X : M \to TM$ be a rough vector field. If $(U, (x^i))$ is any smooth coordinate chart on M, then the restriction of X to U is smooth if and only if its component functions with respect to this chart are smooth.

Proposition 7.1.4

Let M be a smooth manifold with or without boundary, and let $X : M \to TM$ be a rough vector field. The following are equivalent:

(a) X is smooth.
(b) For every $f \in C^\infty(M)$, the function Xf is smooth on M.
(c) For every open subset $U \subseteq M$ and every $f \in C^\infty(U)$, the function Xf is smooth on U.

Definition 7.1.5 ▶ Vector Field as a Derivation

A smooth vector field $X \in \mathfrak{X}(M)$ defines a map

$$\begin{aligned} X : C^\infty(M) &\to C^\infty(M) \\ f &\mapsto Xf \end{aligned}$$

which is linear over $\mathbb{R}$ and following the product rule for vector fields

$$X(fg) = fXg + gXf$$

We call this map a ***Derivation on*** $C^\infty(M)$.

Proposition 7.1.6

Let M be a smooth manifold with or without boundary. A map $D : C^\infty(M) \to C^\infty(M)$ is a derivation if and only if it is of the form $Df = Xf$ for some smooth vector field $X \in \mathfrak{X}(M)$.

Proposition 7.1.7

Suppose M and N are smooth manifolds with or without boundary, and $F : M \to N$ is a diffeomorphism. For every $X \in \mathfrak{X}(M)$, there is a unique smooth vector field on N that is F-related to X.

7.2 Vector Field and Smooth Maps

Definition 7.2.1 ▶ F -related vector fields

Suppose $F : M \to N$ is smooth and X is a vector field on M, and suppose there happens to be a vector field Y on N with the property that for each $p \in M$, $dF_p(X_p) = Y_{F(p)}$. In this case, we say the vector fields X and Y are F-***related***.

Remark.
这是 F-相关向量场的逐点定义.

Proposition 7.2.2

Suppose $F : M \to N$ is a smooth map between manifolds with or without boundary, $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$. Then X and Y are F-related if and only if for every smooth real-vealued function f defined on an open subset of N,

$$X(f \circ F) = (Yf) \circ F$$

Remark.
- 这是 F-相关向量场的 $C^\infty_{F(p)}(M)$-导子刻画.
- F 参与了以下活动: 将函数芽 $f \in C^\infty_{F(p)}(N)$ 的定义域拉回到 M 上以便于 X 作用, 以及将 (Yf) 的定义域 (Y 的取值) 拉回到 M 上以便于比较.

Proof. For any $p \in M$ and any $f \in C^\infty_{F(p)}(M)$,

$$X(f \circ F)(p) = X_p(f \circ F) = \mathrm{d}F_p(X_p) f$$

while

$$(Yf) \circ F(p) = (Yf)(F(p)) = Y_{F(p)} f$$

□

8 Lie Groups

8.1 Basic Definitions

Definition 8.1.1

A ***Lie group*** is a smooth manifold G (without boundary) that is also a group in the algebraic sense, with the property that the multiplication map $m\colon\ G \times G \to G$ and inversion map $i\colon\ G \to G$, given by

$$m(g, h) = gh, \qquad i(g) = g^{-1},$$

are both smooth.

Definition 8.1.2

If G is a Lie group, any element $g \in G$ defines maps $L_g, R_g : G \to G$, called ***left translation*** and ***right translation***, respectively, by

$$L_g(h) = gh, \qquad R_g(h) = hg.$$

which are diffeomorphisms.

Proof. Because L_g can be expressed as the composition of smooth maps

$$G \xrightarrow{\iota_g} G \times G \xrightarrow{m} G,$$

where $\iota_g(h) = (g, h)$ and m is multiplication, it follows that L_g is smooth. It is actually a diffeomorphism of G, because $L_{g^{-1}}$ is a smooth inverse for it. Similarly, $R_g : G \to G$ is a diffeomorphism. $\square$

8.2 Lie Subgroups

Definition 8.2.1

Suppose G is a Lie group. A ***Lie subgroup of*** G is a subgroup of G endowed with a topology and smooth structure making it into a Lie group and an ***immersed submanifold*** of G.

Remark.

- A Lie subgroup of a compact Lie group need not be compact.

Proposition 8.2.2

Suppose G is a Lie group, and $W \subseteq G$ is any neighborhood of the identity.

1. W generates an open subgroup of G.
2. If W is connected, it generates a connected open subgroup of G.
3. If G is connected, then W generates G.

Note.

S 生成子群的构造性表述，就是 S 及 S^{-1} 的有限积的全体.

8.3 Group Actions and Equivariant Maps

Definition 8.3.1

- If G is a group and M is a set, a ***left action of*** G ***on*** M is a map $G \times M \to M$, often written as $(g, p) \mapsto g \cdot p$, that satisfies

$$g_1 \cdot (g_2 \cdot p) = (g_1 g_2) \cdot p \quad \text{for all } g_1, g_2 \in G \text{ and } p \in M;$$
$$e \cdot p = p \quad \text{for all } p \in M. \tag{8.1}$$

- A ***right action*** is defined analogously as a map $M \times G \to M$ with the appropriate composition law:

$$(p \cdot g_1) \cdot g_2 = p \cdot (g_1 g_2) \quad \text{for all } g_1, g_2 \in G \text{ and } p \in M;$$
$$p \cdot e = p \quad \text{for all } p \in M.$$

- If M is a topological space and G is a topological group, an action of G on M is said to be a ***continuous action*** if the defining map

$G \times M \to M$ or $M \times G \to M$ is continuous. In this case, M is said to be a ***(left or right)*** ***G-space***.

- If in addition M is a smooth manifold with or without boundary, G is a Lie group, and the defining map is smooth, then the action is said to be a ***smooth action***.

Definition 8.3.2

Sometimes it is useful to give a name to an action, such as $\theta : G \times M \to M$, with the action of a group element g on a point p usually written as $\theta_g(p)$.

- In terms of this notation, the conditions for a left action read

$$\theta_{g_1} \circ \theta_{g_2} = \theta_{g_1 g_2}, \qquad \theta_e = \mathrm{Id}_M, \tag{8.2}$$

- while for a right action the first equation is replaced by

$$\theta_{g_2} \circ \theta_{g_1} = \theta_{g_1 g_2}. \tag{8.3}$$

- For a smooth action, each map $\theta_g : M \to M$ is a diffeomorphism, because $\theta_{g^{-1}}$ is a smooth inverse for it.

Example 8.3.3

Every Lie Group G acts smoothly on itself by translation.

Proof. Given any two points $g_1, g_2 \in G$, there is a unique left translation of G taking g_1 to g_2, namely left translation by $g_2 g_1^{-1}$; thus the action is both free and transitive. □

Definition 8.3.4

Suppose $\theta : G \times M \to M$ is a left action of a group G on a set M.

- For each $p \in M$, the ***orbit of*** p, denoted by $G \cdot p$, is the set of all images of p under the action by elements of G:

$$G \cdot p = \{g \cdot p : g \in G\}.$$

- For each $p \in M$, the ***isotropy group*** or ***stabilizer of*** p, denoted by

G_p, is the set of elements of G that fix p:

$$G_p = \{g \in G : g \cdot p = p\}.$$

The definition of a group action guarantees that G_p is a subgroup of G.

- The action is said to be ***transitive*** if for every pair of points $p, q \in M$, there exists $g \in G$ such that $g \cdot p = q$, or equivalently if the only orbit is all of M.
- The action is said to be ***free*** if the only element of G that fixes any element of M is the identity: $g \cdot p = p$ for some $p \in M$ implies $g = e$, or equivalently if every isotropy group is trivial.

8.4 Lie Bracket

Definition 8.4.1 ▸ Lie Bracket

We define an operator $[X, Y] : C^\infty(M) \to C^\infty(M)$, called the ***Lie bracket of*** X ***and*** Y , by

$$[X, Y] f = XY f - YX f$$

Remark.
The Lie bracket of any pair of smooth vector fields is a smooth vector field.

Proposition 8.4.2 ▸ Coordinate Formula for the Lie Bracket

Let X, Y be smooth vector fields on a smooth manifold M with or without boundary, and let $X = X^i \frac{\partial}{\partial x^i}$ and $Y = Y^j \frac{\partial}{\partial x^j}$ be the coordinate expressions for X and Y in terms of some smooth local coordinates (x^i) for M. Then $[X, Y]$ has the following coordinate expression:

$$[X, Y] = \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right) \frac{\partial}{\partial x^j}, \tag{8.4}$$

or more concisely,

$$[X, Y] = \left(XY^j - YX^j\right)\frac{\partial}{\partial x^j}. \tag{8.5}$$

Proposition 8.4.3 ▶ Naturality of the Lie Bracket

Let $F : M \to N$ be a smooth map between manifolds with or without boundary, and let $X_1, X_2 \in \mathfrak{X}(M)$ and $Y_1, Y_2 \in \mathfrak{X}(N)$ be vector fields such that X_i is F-related to Y_i for $i = 1, 2$. Then $[X_1, X_2]$ is F-related to $[Y_1, Y_2]$.

Proof. □

8.5 The Lie Algebra of a Lie Group

Definition 8.5.1

Suppose G is a Lie group. Recall that G acts smoothly and transitively on itself by left translation: $L_g(h) = gh$. A vector field X on G is said to be ***left-invariant*** if it is invariant under all left translations, in the sense that it is L_g-related to itself for every $g \in G$. More explicitly, this means

$$d(L_g)_{g'}(X_{g'}) = X_{gg'}, \quad \text{for all } g, g' \in G.$$

Since L_g is a diffeomorphism, this can be abbreviated by writing $(L_g)_*X = X$ for every $g \in G$.

Remark.

- Because $\left(L_g\right)_*(aX + bY) = a\left(L_g\right)_* X + b\left(L_g\right)_* Y$, the set of all smooth left-invariant vector fields on G is a linear subspace of $\mathfrak{X}(G)$.

Proposition 8.5.2

Let G be a Lie group, and suppose X and Y are smooth left-invariant vector fields on G. Then $[X, Y]$ is also left-invariant.

Proof. Let $g \in G$ be arbitrary. Since X, Y is L_g-related to itself. We have from the naturality of the Lie bracket that $[X, Y]$ is L_g-related to $\left[\left(L_g\right)_* X, \left(L_g\right)_* Y\right] = [X, Y]$. □

Corollary 8.5.3

The collection of all smooth left-invariant vector fields on a Lie group G is a Lie algebra under the Lie bracket, called the ***Lie algebra of*** G, and is denoted by $\operatorname{Lie}(G)$.

Theorem 8.5.4

Let G be a Lie group. The evaluation map $\varepsilon\colon \operatorname{Lie}(G) \to T_eG$, given by $\varepsilon(X) = X_e$, is a vector space isomorphism. Thus, $\operatorname{Lie}(G)$ is finite-dimensional, with dimension equal to $\dim G$.

Proof sketch.
左不变向量场的全体是线性空间, ε 是线性映射.
按照左不变性将 $v \in T_eG$ 搬到全体 G 上, 得到一个唯一的左不变向量场.

Proof. It is clear from the definition that ε is linear over $\mathbb{R}$. It is easy to prove that it is injective: if $\varepsilon(X) = X_e = 0$ for some $X \in \operatorname{Lie}(G)$, then the left-invariance of X implies that $X_g = \mathrm{d}\left(L_g\right)_e(X_e) = 0$ for every $g \in G$, so $X = 0$.

To show that ε is surjective, let $v \in T_eG$ be arbitrary, and define a (rough) vector field v^L on G by

$$v^L\big|_g = d\left(L_g\right)_e(v)$$

If there is a left-invariant vector filed on G whose value at the identity is v, clearly it has to be given this formula.

First we need to check that v^L is smooth. It suffices to show that $v^L f$ is smooth whenever $f \in C^\infty(G)$. Choose a □

Proposition 8.5.5 ▶ Lie Algebra of the General Linear Group

The composition of the natural maps

$$\operatorname{Lie}\big(\mathrm{GL}(n,\mathbb{R})\big) \to T_{I_n}\mathrm{GL}(n,\mathbb{R}) \to \mathfrak{gl}(n,\mathbb{R})$$

gives a Lie algebra isomorphism between $\operatorname{Lie}\big(\mathrm{GL}(n,\mathbb{R})\big)$ and the matrix algebra $\mathfrak{gl}(n,\mathbb{R})$.

Remark.
The vector spaces $\operatorname{Lie}\big(\mathrm{GL}(n,\mathbb{R})\big)$ and $\mathfrak{gl}(n,\mathbb{R})$ have independently de-

fined Lie algebra structures—the first coming from Lie brackets of vector fields, and the second from commutator brackets of matrices.

Proof sketch.
$T_{I_n}\,\mathrm{GL}(n,\mathbb{R})$ 和 $\mathfrak{gl}(n,\mathbb{R})$ 有自然的同构, 前者有左不变向量场的李括号构成的李代数结构, 后者有矩阵的交换子构成的李代数结构.
$\mathrm{Lie}(\mathrm{GL}(n,\mathbb{R}))$ 有李括号构成的李代数结构, $\mathfrak{gl}(n,\mathbb{R})$ 有矩阵的交换子构成的李代数结构, 我们通过计算说明这两个结构是一致的.
左平移是 $\mathrm{GL}(n,\mathbb{R})$ 上线性映射, 左不变向量场的微分是与其形式相同的线性映射.

Proof. Using the maxtrix entires X^i_j as the coordinates. There is a natural isomorphism

$$A^i_j \frac{\partial}{\partial X^i_j}\bigg|_{I_n} \mapsto \left(A^i_j\right)$$

Then for each matrix $A = \left(A^i_j\right) \in \mathfrak{gl}(n,\mathbb{R})$, A gives a left-invariant vector field $A^L \in \mathfrak{g}$, by

$$A^L|_X = (dL_X)\left(A^i_j \frac{\partial}{\partial X^i_j}\right)$$

The differential of $L_X : A \mapsto XA$ is just the linear map of the same form , that is

$$A^L|_X = X^i_j A^j_k \frac{\partial}{\partial X^i_k}$$

Then

$$\begin{aligned}
&[A^L|_X, B^L|_X] \\
&= \left(X^i_j A^j_k \frac{\partial}{\partial X^i_k}\right)\left(X^p_q B^q_r \frac{\partial}{\partial X^q_r}\right) - \left(X^p_q B^q_r \frac{\partial}{\partial X^q_r}\right)\left(X^i_j A^j_k \frac{\partial}{\partial X^i_k}\right) \\
&= X^i_j A^j_k B^k_r \frac{\partial}{\partial X^i_r} - X^p_q B^q_r A^r_k \frac{\partial}{\partial X^q_k} \\
&= X^i_j A^j_k B^k_r \frac{\partial}{\partial X^i_r} - X^j_i B^i_k A^k_r \frac{\partial}{\partial X^i_r} \\
&= \left(X^i_j A^j_k B^k_r - X^j_i B^i_k A^k_r\right) \frac{\partial}{\partial X^i_r}
\end{aligned}$$

Evaluate at I_n, we get

$$[A^L|_X, B^L|_X]_{I_n} = \left(A^i_k B^k_r - B^i_k A^k_r\right) \frac{\partial}{\partial X^i_r} = (AB - BA)^i_r \frac{\partial}{\partial X^i_r}$$

□

Induced Lie Algebra Homomorphisms

Theorem 8.5.6 ▶ Induced Lie Algebra Homomorphisms

Let G and H be Lie groups, and let $\mathfrak{g}$ and $\mathfrak{h}$ be their Lie algebras. Suppose $F : G \to H$ is a Lie group homomorphism. For every $X \in \mathfrak{g}$, there is a unique vector field in $\mathfrak{h}$ that is F-related to X. With this vector field denoted by F_*X, the map $F_* : \mathfrak{g} \to \mathfrak{h}$ so defined is a Lie algebra homomorphism, is called the ***induced Lie algebra homomorphism***.

Remark.
Note that the theorem implies that for any left-invariant vector field $X \in \mathfrak{g}$, F_*X is a well-defined smooth vector field on H, even though F may not be a diffeomorphism.

Proof sketch.
- 左不变性使得相关向量场的选取是唯一且确定的，唯一需要验证的是相关性.
- 群同态保证了关于平移的自然性，$F \circ L_g = L_{F(g)} \circ F$. 它的微分形式 $dF \circ d(L_g) = d(L_{F(g)}) \circ dF$. 既是左不变向量场之间的相关性.
- F_* 的 Lie 代数同态性，来自于李括号的相关性 $F_*[X, Y] = [F_*X, F_*Y]$.

8.6 One-Parameter Subgroups and the Exponential Map

8.6.1 One-Parameter Subgroups

Definition 8.6.1

A ***one-parameter subgroup of*** G is defined to be a Lie group homomorphism $\gamma : \mathbb{R} \to G$, with $\mathbb{R}$ considered as a Lie group under addition.

Theorem 8.6.2 ▸ Characterization of One-Parameter Subgroups

Let G be a Lie group. The one-parameter subgroups of G are precisely the maximal integral curves of left-invariant vector fields starting at the identity.

Remark.
Given $X \in \mathrm{Lie}(G)$, the one-parameter subgroup determined by X in this way is called the ***one-parameter subgroup generated by*** X. Because left-invariant vector fields are uniquely determined by their values at the identity, it follows that each one-parameter subgroup is uniquely determined by its initial velocity in T_eG, and thus there are one-to-one correspondences

$$\{\text{one-parameter subgroups of } G\} \longleftrightarrow \mathrm{Lie}(G) \longleftrightarrow T_eG.$$

Proof sketch.
$\Longleftarrow$：给定积分曲线 γ, 为了证明群律 $\gamma(s+t) = \gamma(s)\gamma(t)$，以重参数化和左平移两种方式将积分曲线的起点搬到 $\gamma(s)$, 其中后者依赖左不变向量场的平移不变形.
$\Longrightarrow$：群同态 γ 给出李代数同态 γ_*, $\gamma_*\left(\frac{\mathrm{d}}{\mathrm{d}t}\right)$ 就是一个左不变向量场, 它恰好是 γ 的切向量场.

Proof. Supposet that γ is a maximal integral curve of a left-invariant vector field X. Since X is complete, γ has the domain $\mathbb{R}$. The left-invariance means that X is L_g-invariant for each $g \in G$, then L_g take the integral curve of X to the integral curve of X. $t \mapsto L_{\gamma(s)} \circ \gamma(t)$ is the integral curve of X starting at $\gamma(s)$. The rescaling lemma shows that $t \mapsto \gamma(s+t)$ is also the integral curve of X starting at $\gamma(s)$, then it follows from the uniqueness of the integral curve that

$$\gamma(s+t) = L_{\gamma(s)} \circ \gamma(t) = \gamma(s)\gamma(t)$$

Conversely , if γ is the one-parameter subgroup of of G.Let $X = \gamma_*\left(\frac{\mathrm{d}}{\mathrm{d}t}\right) \in \mathrm{Lie}(G)$, treating $\frac{\mathrm{d}}{\mathrm{d}t}$ as a left-invariant vector filed on $\mathbb{R}$. □

Proposition 8.6.3

Suppose G is a Lie group and $H \subseteq G$ is a Lie subgroup. The one-parameter subgroups of H are precisely those one-parameter subgroups of G whose initial velocities lie in T_eH.

Proof sketch.
单参数子群与原点的切空间是对应的.
单参数子群就是过原点的积分曲线, 只要初速度一样, 它们就是相同的积分曲线.

8.6.2 The Exponential Map

Definition 8.6.4 ▶ Exponential map

Given a Lie group G with Lie algebra $\mathfrak{g}$, we define a map $\exp : \mathfrak{g} \to G$, called the ***exponential map of*** G, as follows: for any $X \in \mathfrak{g}$, we set

$$\exp X = \gamma(1),$$

where γ is the one-parameter subgroup generated by X, or equivalently the integral curve of X starting at the identity .

Proposition 8.6.5 ▶ Properties of the Exponential Map

Let G be a Lie group and let $\mathfrak{g}$ be its Lie algebra.

(a) The exponential map is a smooth map from $\mathfrak{g}$ to G.
(b) For any $X \in \mathfrak{g}$ and $s, t \in \mathbb{R}$, $\exp(s+t)X = \exp sX \exp tX$.
(c) For any $X \in \mathfrak{g}$, $(\exp X)^{-1} = \exp(-X)$.
(d) For any $X \in \mathfrak{g}$ and $n \in \mathbb{Z}$, $(\exp X)^n = \exp(nX)$.
(e) The differential $(\mathrm{d}\exp)_0 : T_0\mathfrak{g} \to T_eG$ is the identity map, under the canonical identifications of both $T_0\mathfrak{g}$ and T_eG with $\mathfrak{g}$ itself.
(f) The exponential map restricts to a diffeomorphism from some

neighborhood of 0 in $\mathfrak{g}$ to a neighborhood of e in G.

(g) If H is another Lie group, $\mathfrak{h}$ is its Lie algebra, and $\Phi : G \to H$ is a Lie group homomorphism, the following diagram commutes:

$$\begin{tikzcd} \mathfrak{g} \arrow[r, "\Phi_*"] \arrow[d, "\exp"'] & \mathfrak{h} \arrow[d, "\exp"] \\ G \arrow[r, "\Phi"'] & H. \end{tikzcd} \tag{8.6}$$

(h) The flow θ of a left-invariant vector field X is given by $\theta_t = R_{\exp tX}$ (right multiplication by $\exp tX$).

Proposition 8.6.6 ▸

Let G be a Lie group. For any $X \in \mathrm{Lie}(G)$, $\gamma(s) = \exp sX$ is the one-parameter subgroup of G generated by X.

Proof. Let $\gamma : \mathbb{R} \to G$ be the one-parameter subgroup generated by X, which is the integral curve of X starting at e. For any fixed $s \in \mathbb{R}$, the rescaling lemma of integral curve gives that $\tilde{\gamma}(t) = \gamma(st)$ is the integral cruve of sX starting at e, so

$$\exp sX = \tilde{\gamma}(1) = \gamma(s)$$

□

Proposition 8.6.7

et G be a Lie group, and let $H \subseteq G$ be a Lie subgroup. With $\mathrm{Lie}(H)$ considered as a subalgebra of $\mathrm{Lie}(G)$ in the usual way, the exponential map of H is the restriction to $\mathrm{Lie}(H)$ of the exponential map of G, and

$$\mathrm{Lie}(H) = \{X \in \mathrm{Lie}(G) : \exp tX \in H \text{ for all } t \in \mathbb{R}\}.$$

8.7 The Closed Subgroup Theorem

Theorem 8.7.1 ▸ Closed Subgroup Theorem

Suppose G is a Lie group and $H \subseteq G$ is a subgroup that is also a closed subset of G. Then H is an embedded Lie subgroup.

8.8 The Lie Correspondence

Theorem 8.8.1 ▸

Suppose G and H are Lie groups with G simply connected, and let $\mathfrak{g}$ and $\mathfrak{h}$ be their Lie algebras. For any Lie algebra homomorphism $\varphi : \mathfrak{g} \to \mathfrak{h}$, there is a unique Lie group homomorphism $\Phi : G \to H$ such that $\Phi_* = \varphi$.

Corollary 8.8.2

If G and H are simply connected Lie groups with isomorphic Lie algebras, then G and H are isomorphic.

Theorem 8.8.3 ▸ The Lie Correspondence

There is a one-to-one correspondence between isomorphism classes of finite-dimensional Lie algebras and isomorphism classes of simply connected Lie groups, given by associating each simply connected Lie group with its Lie algebra.

8.9 Normal Subgroups

Lemma 8.9.1

Let G be a connected Lie group, and let $H \subseteq G$ be a connected Lie subgroup. Let $\mathfrak{g}$ and $\mathfrak{h}$ denote the Lie algebras of G and H, respectively. Then H is normal in G if and only if

$$(\exp X)(\exp Y)(\exp(-X)) \in H \quad \text{for all } X \in \mathfrak{g} \text{ and } Y \in \mathfrak{h}. \tag{8.7}$$

Proof sketch.
exp 是 $0 \in \mathfrak{h}$ 到 $e \in H$ 的局部的微分同胚, 也是 $0 \in \mathfrak{g}$ 到 $e \in G$ 的微分同胚. 于是 $e \in H$ 和 $e \in G$ 分别有一个以 exp 像构成的开邻域 U, U_0. 而连通李群的单位元的开邻域生成整个李群, 故 U 和 U_0 分别生成了 H 和 G.
题设条件应用在这两个 exp 像构成的开邻域上, 给出 U 中元素在 U_0 从元素的共轭下落在 H 中. 利用生成的性质 (H 和 G 中的元素分别写成 U 和 U_0 中的元素的有限积). 我们可以将 ghg^{-1} 拆分成若干 $g_i h_j g_i^{-1}$ 的组合, 其中 $g_i \in U_0, h_j \in U$.

8.9.1 The Adjoint Representation

Definition 8.9.2

Let G be a Lie group and $\mathfrak{g}$ be its Lie algebra. For any $g \in G$, the conjugation map $C_g : G \to G$ given by $C_g(h) = ghg^{-1}$ is a Lie group homomorphism. We let $\mathrm{Ad}(g) = (C_g)_* : \mathfrak{g} \to \mathfrak{g}$ denote its induced Lie algebra homomorphism.

Proposition 8.9.3 ▶ The Adjoint Representation

If G is a Lie group with Lie algebra $\mathfrak{g}$, the map $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ is a Lie group representation, called the ***adjoint representation*** of G.

Definition 8.9.4

Given a finite dimensional Lie algebra $\mathfrak{g}$, for each $X \in \mathfrak{g}$, define a map $\mathrm{ad}(X) : \mathfrak{g} \to \mathfrak{g}$ by $\mathrm{ad}(X)Y = [X, Y]$.

Proposition 8.9.5 ▶ adjoint representation of $\mathfrak{g}$

For any Lie algebra $\mathfrak{g}$, the map $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ is a Lie algebra representation, called the ***adjoint representation of*** $\mathfrak{g}$.

Theorem 8.9.6

Let G be a Lie group, let $\mathfrak{g}$ be its Lie algebra, and let $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ be the adjoint representation of G. The induced Lie algebra representation $\mathrm{Ad}_* : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ is given by $\mathrm{Ad}_* = \mathrm{ad}$.

9

Integral Curves and Flows

9.1 Integral Curves

Definition 9.1.1 ▸ Integral Curve

If V is a vector field on M, an ***integral curve of*** V is a differentiable curve $\gamma : J \to M$ whose velocity at each point is equal to the value of V at that point:

$$\gamma'(t) = V_{\gamma(t)} \quad \text{for all } t \in J.$$

If $0 \in J$, the point $\gamma(0)$ is called the ***starting point of*** γ.

Finding integral curves boils down to sloving a system of ordinary differential equations in a smooth chat. Suppose V is a smooth vector field on M and $\gamma : J \to M$ is a smooth curve. On a smooth coordinate domain $U \subseteq M$, we can write γ in local coordinates as

$$\gamma(t) = \left(\gamma^1(t), \cdots, \gamma^n(t)\right)$$

Then the condition $\gamma'(t) = V_{\gamma(t)}$ for γ to be an integral curve of V can be written

$$\dot{\gamma}^i(t) \left.\frac{\partial}{\partial x^i}\right|_{\gamma(t)} = V^i(\gamma(t)) \left.\frac{\partial}{\partial x^i}\right|_{\gamma(t)},$$

which reduces to the following autonomous system of ordinary differential equatoins:

$$\begin{aligned} \dot{\gamma}^1(t) &= V^1(\gamma^1(t), \dots, \gamma^n(t)), \\ &\vdots \\ \dot{\gamma}^n(t) &= V^n(\gamma^1(t), \dots, \gamma^n(t)). \end{aligned}$$

Proposition 9.1.2

Let V be a smooth vector field on a smooth manifold M. For each point $p \in M$, there exists $\varepsilon > 0$ and a smooth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ that is an integral curve of V starting at p.

Proof. The ODE system corosponding to the condition

$$\gamma'(t) = V_{\gamma(t)}$$

is smooth. Thus it satisfies the local Lipschitz condition. Then the proposition follows from the Picard-Lindelöf Theorem. ☐

Proposition 9.1.3 ▸ Naturality of Integral Curves

Suppose M and N are smooth manifolds and $F : M \to N$ is a smooth map. Then $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are F-related if and only if F takes integral curves of X to integral curves of Y, meaning that for each integral curve γ of X, $F \circ \gamma$ is an integral curve of Y.

Proof. Suppose that X and Y are F-related, and $\gamma : J \to M$ is an integral curve of X. If we define $\sigma : J \to N$ by $\sigma = F \circ \gamma$, then

$$\sigma'(t) = dF_{\gamma(t)}\gamma'(t) = dF_{\gamma(t)}X_{\gamma(t)} = Y_{F(\gamma(t))} = Y_{\sigma(t)}$$

so σ is an integral curve of Y.

Conversely, suppose F take integral curves of X to integral curves of Y. Given $p \in M$, let $\gamma : (-\varepsilon, \varepsilon) \to M$ be an integral curve of X starting at p. Since $F \circ \gamma$ is an integral curve of Y starting at $F(p)$, we have

$$Y_{F(p)} = (F \circ \gamma)'(0) = \mathrm{d}F_p(\gamma'(0)) = dF_p\left(X_p\right)$$

which shows that X and Y are F-related. ☐

9.2 Flows

Definition 9.2.1

We define a ***global flow*** on M (also called a ***one-parameter group action***) to be a continuous left $\mathbb{R}$-action on M; that is, a continuous map $\theta : \mathbb{R} \times M \to M$ satisfying the following properties for all $s, t \in \mathbb{R}$ and $p \in M$:

$$\theta(t, \theta(s, p)) = \theta(t + s, p), \quad \theta(0, p) = p. \tag{9.1}$$

Given a global flow θ on M, we define two collections of maps as follows:

- For each $t \in \mathbb{R}$, define a continuous map $\theta_t : M \to M$ by

$$\theta_t(p) = \theta(t, p).$$

The defining properties (9.1) are equivalent to the ***group laws***

$$\theta_t \circ \theta_s = \theta_{t+s}, \quad \theta_0 = \mathrm{Id}_M . \tag{9.2}$$

As is the case for any continuous group action, each map $\theta_t : M \to M$ is a homeomorphism, and if the flow is smooth, θ_t is a diffeomorphism.
- For each $p \in M$, define a curve $\theta^{(p)} : \mathbb{R} \to M$ by

$$\theta^{(p)}(t) = \theta(t, p).$$

The image of this curve is the orbit of p under the group action.

Proposition 9.2.2

If $\theta : \mathbb{R} \times M \to M$ is a smooth global flow, for each $p \in M$ we define a tangent vector $V_p \in T_pM$ by

$$V_p = \theta^{(p)\prime}(0).$$

The assignment $p \mapsto V_p$ is a (rough) vector field on M, which is called the ***infinitesimal generator of*** θ.
The infinitesimal generator V of θ is a smooth vector field on M, and each curve $\theta^{(p)}$ is an integral curve of V.

9.2.1 Fundemental Theorem on Flows

Definition 9.2.3

- If M is a manifold, a ***flow domain*** for M is an open subset $\mathcal{D} \subseteq \mathbb{R} \times M$ with the property that for each $p \in M$, the set $\mathcal{D}^{(p)} = \{t \in \mathbb{R} : (t, p) \in \mathcal{D}\}$ is an open interval containing 0 .
- A ***flow*** on M is a continuous map $\theta : \mathcal{D} \to M$, where $\mathcal{D} \subseteq \mathbb{R} \times M$ is a flow domain, that satisfies the following group laws :
  - For all $p \in M$,

$$\theta(0, p) = p, \tag{9.3}$$

– and for all $s \in \mathcal{D}^{(p)}$ and $t \in \mathcal{D}^{(\theta(s,p))}$ such that $s + t \in \mathcal{D}^{(p)}$,

$$\theta(t, \theta(s, p)) = \theta(t + s, p). \tag{9.4}$$

We sometimes call θ a ***local flow*** to distinguish it from a global flow as defined earlier. The unwieldy term ***local one-parameter group action*** is also used.

Note.

- ***flows domain*** 要求每一个空间节点上，时间的变化范围都是包含 0 的开区间
- 一个 ***flow*** 就是 M 上活动在 $\mathcal{D}$ 内关于时间变化的一个局部左 $\mathbb{R}$-作用.

Definition 9.2.4

If θ is a flow, we define $\theta_t(p) = \theta^{(p)}(t) = \theta(t, p)$ whenever $(t, p) \in \mathcal{D}$, just as for a global flow. For each $t \in \mathbb{R}$, we also define

$$M_t = \{p \in M : (t, p) \in \mathcal{D}\}, \tag{9.5}$$

so that

$$p \in M_t \Leftrightarrow t \in \mathcal{D}^{(p)} \Leftrightarrow (t, p) \in \mathcal{D}.$$

If θ is smooth, the ***infinitesimal generator of*** θ is defined by $V_p = \theta^{(p)\prime}(0)$.

Note.

- 当我们谈论点在 t 时间后的行为时，只能对 M_t 上的点讨论.
- 无穷小生成元就是每一点处运动曲线的起始切向量.

Theorem 9.2.5 ▶ Fundamental Theorem on Flows

Let V be a smooth vector field on a smooth manifold M. There is a unique smooth maximal flow $\theta : \mathcal{D} \to M$ whose infinitesimal generator is V. This flow has the following properties:

(a) For each $p \in M$, the curve $\theta^{(p)} : \mathcal{D}^{(p)} \to M$ is the unique maximal integral curve of V starting at p.

(b) If $s \in \mathcal{D}^{(p)}$, then $\mathcal{D}^{(\theta(s,p))}$ is the interval $\mathcal{D}^{(p)} - s = \{t - s : t \in \mathcal{D}^{(p)}\}$.

(c) For each $t \in \mathbb{R}$, the set M_t is open in M, and $\theta_t : M_t \to M_{-t}$ is a diffeomorphism with inverse θ_{-t}.

Note.

- 向量场 V 的最大流 $\theta : \mathcal{D} \subset \mathbb{R} \times M \to M$ 将从 M 中各点出发的极大积分曲线统一表示为 $\gamma_p(t) = \theta(t, p)$. 相对于单纯的一族积分曲线, 最大流 θ 还同时编码了解对初值 p 的光滑依赖, 即 θ 作为 (t, p) 的函数是光滑的
- 直观上, [(c)] 用平移来看. 划定一个区域, 将这些区域整体平移, 删去那些跑到区域外的点, 删去的点正好能补齐因为平移而产生的空缺, 这体现了平移关于时间的可逆性, 平移就是一种特殊的流.

9.2.2 Complete Vector Fields

Definition 9.2.6

We say that a smooth vector field is ***complete*** if it generates a global flow, or equivalently if each of its maximal integral curves is defined for all $t \in \mathbb{R}$.

Lemma 9.2.7 ▸ Uniform Time Lemma

Let V be a smooth vector field on a smooth manifold M, and let θ be its flow. Suppose there is a positive number ε such that for every $p \in M$, the domain of $\theta^{(p)}$ contains $(-\varepsilon, \varepsilon)$. Then V is complete.

Proof. Suppose for the sake of contradiction for some $p \in M$, the domain $\mathcal{D}^{(p)}$ of $\theta^{(p)}$ is bounded above. Let $b = \sup \mathcal{D}^{(p)}$, let t_0 be a positive number such that $b - \varepsilon < t_0 < b$, and let $q = \theta^{(p)}(t_0)$. The hypothesis implies that $\theta^{(q)}(t)$ is defined at least for $t \in (-\varepsilon, \varepsilon)$. Define curve $\gamma : (-\varepsilon, t_0 + \varepsilon) \to M$ by

$$\gamma(t) = \begin{cases} \theta^{(p)}(t), & -\varepsilon < t < b \\ \theta^{(q)}(t - t_0), & t_0 - \varepsilon < t < t_0 + \varepsilon \end{cases}$$

These two definitions agree where they overlap, because $\theta^{(q)}(t - t_0) = \theta_{t-t_0}(q) = \theta_{t-t_0}\theta_{t_0}(p) = \theta_t(p) = \theta^{(p)}(t)$ by the group law for θ. γ is an integral curve starting at p. Since $t_0 je\varepsilon > b$, this is a contradiction. $\square$

9.3 Lie Derivatives

Definition 9.3.1

Suppose M is a smooth manifold, V is a smooth vector field on M, and θ is the flow of V. For any smooth vector field W on M, define a rough vector field on M, denoted by $\mathcal{L}_V W$ and called the ***Lie derivative of W with respect to V***, by

$$(\mathcal{L}_V W)_p = \left.\frac{d}{dt}\right|_{t=0} d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) = \lim_{t\to 0} \frac{d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) - W_p}{t}, \tag{9.6}$$

provided the derivative exists. For small $t \neq 0$, at least the difference quotient makes sense: θ_t is defined in a neighborhood of p, and θ_{-t} is the inverse of θ_t, so both $d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)})$ and W_p are elements of T_pM
.

Theorem 9.3.2

If M is a smooth manifold and $V, W \in \mathfrak{X}(M)$, then $\mathcal{L}_V W = [V, W]$.

9.4 Commuting Vector Fields

Definition 9.4.1

If θ is a smooth flow, a vector field W is said to be ***invariant under*** θ if W is θ_t-related to itself for each t; more precisely, this means that $W|_{M_t}$ is θ_t-related to $W|_{M_{-t}}$ for each t, or equivalently that $d(\theta_t)_p(W_p) = W_{\theta_t(p)}$ for all (t, p) in the domain of θ.

Theorem 9.4.2

For smooth vector fields V and W on a smooth manifold M, the following are equivalent:

(a) V and W commute.
(b) W is ***invariant under the flow*** of V.
(c) V is ***invariant under the flow*** of W.

Theorem 9.4.3 ▸ Canonical Form for Commuting Vector Fields

Let M be a smooth n-manifold, and let $(V_1, \dots, V_k)$ be a linearly independent k-tuple of smooth commuting vector fields on an open subset $W \subseteq M$.

- For each $p \in W$, there exists a smooth coordinate chart $(U, (s^i))$ centered at p such that $V_i = \partial/\partial s^i$ for $i = 1, \dots, k$.
- If $S \subseteq W$ is an embedded codimension-k submanifold and p is a point of S such that T_pS is complementary to the span of $(V_1|_p, \dots, V_k|_p)$, then the coordinates can also be chosen such that $S \cap U$ is the slice defined by $s^1 = \cdots = s^k = 0$.

10

Distributions and Foliations

10.1 Distributions and Involutivity

Definition 10.1.1

Let M be a smooth manifold.

- A ***distribution on M of rank k*** is a rank-k subbundle of TM. It is called a ***smooth distribution*** if it is a smooth subbundle.
- Often a rank-k distribution is described by specifying for each $p \in M$ a linear subspace $D_p \subseteq T_pM$ of dimension k, and letting $D = \bigcup_{p \in M} D_p$. It then follows from the local frame criterion for subbundles (Lemma 10.32) that D is a smooth distribution if and only if each point of M has a neighborhood U on which there are smooth vector fields $X_1, \dots, X_k : U \to TM$ such that $X_1|_q, \dots, X_k|_q$ form a basis for D_q at each $q \in U$. In this case, we say that D is the distribution (locally) ***spanned by the vector fields*** $X_1, \dots, X_k$.

10.1.1 Integral Manifolds and Involutivity

Definition 10.1.2

Suppose $D \subseteq TM$ is a smooth distribution. A nonempty immersed submanifold $N \subseteq M$ is called an ***integral manifold of D*** if $T_pN = D_p$ at each point $p \in N$.

Definition 10.1.3

Suppose D is a smooth distribution on M. We say that D is ***involutive*** if given any pair of smooth local sections of D [a], their Lie bracket is also a local section of D.

[a](i.e., smooth vector fields X, Y defined on an open subset of M such that $X_p, Y_p \in D_p$ for each p)

10.2 The Frobenius Theorem

Definition 10.2.1

Given a rank-k distribution $D \subseteq TM$, let us say that a smooth coordinate chart (U, φ) on M is ***flat for D*** if $\varphi(U)$ is a cube in $\mathbb{R}^n$, and at points of U, D is spanned by the first k coordinate vector fields $\partial/\partial x^1, \ldots, \partial/\partial x^k$.

Definition 10.2.2

We say that a distribution $D \subseteq TM$ is ***completely integrable*** if there exists a flat chart for D in a neighborhood of each point of M. Obviously, every completely

Theorem 10.2.3 ▸ Frobenius

Every ***involutive distribution*** is ***completely integrable***.

10.3 Foliations

Definition 10.3.1

Let M be a smooth n-manifold, and let $\mathscr{F}$ be any collection of k-dimensional submanifolds of M.

- A smooth chart (U, φ) for M is said to be ***flat for*** $\mathcal{F}$ if $\varphi(U)$ is a cube in $\mathbb{R}^n$, and each submanifold in $\mathcal{F}$ intersects U in either the empty set or a countable union of k-dimensional slices of the form $x^{k+1} = c^{k+1}, \ldots, x^n = c^n$.
- We define a ***foliation of dimension k on M*** to be a collection $\mathcal{F}$ of disjoint, connected, nonempty, immersed k-dimensional submanifolds of M (called the ***leaves of the foliation***), whose union is M, and such that in a neighborhood of each point $p \in M$ there exists a flat chart for $\mathcal{F}$.

11 Quotient Manifolds

11.1 Quotients of Manifolds by Group Actions

11.2 Homogeneous Spaces

Definition 11.2.1

A smooth manifold endowed with a transitive smooth action by a Lie group G is called a ***homogeneous*** G***-space*** .

Theorem 11.2.2 ▶ Homogeneous Space Construction Theorem

Let G be a Lie group and let H be a closed subgroup of G. The left coset space G/H is a topological manifold of dimension equal to $\dim G - \dim H$, and has a unique smooth structure such that the quotient map $\pi : G \to G/H$ is a smooth submersion. The left action of G on G/H given by

$$g_1 \cdot (g_2 H) = (g_1 g_2)H \tag{21.3}$$

turns G/H into a homogeneous G-space.

Theorem 11.2.3 ▶ Homogeneous Space Characterization Theorem

Let G be a Lie group, let M be a homogeneous G-space, and let p be any point of M. The isotropy group G_p is a closed subgroup of G, and the map $F : G/G_p \to M$ defined by $F(gG_p) = g \cdot p$ is an equivariant diffeomorphism.

11.3 Quotient Lie Groups

12 Tensors and Differential Forms

12.1 Multilinear Algebra

Definition 12.1.1

Write $L(V_1, \ldots, V_k; W)$ for the set of all multilinear maps from $V_1 \times \cdots \times V_k$ to W. It is a vector space under the usual operations of pointwise addition and scalar multiplication:

$$(F + F')(v_1, \ldots, v_k) = F(v_1, \ldots, v_k) + F'(v_1, \ldots, v_k),$$

$$(aF)(v_1, \ldots, v_k) = a\big(F(v_1, \ldots, v_k)\big).$$

Definition 12.1.2

Let $V_1, \ldots, V_k, W_1, \ldots, W_l$ be real vector spaces, and suppose $F \in L(V_1, \ldots, V_k; \mathbb{R})$ and $G \in L(W_1, \ldots, W_l; \mathbb{R})$. Define a function

$$F \otimes G : V_1 \times \cdots \times V_k \times W_1 \times \cdots \times W_l \to \mathbb{R}$$

by

$$F \otimes G(v_1, \ldots, v_k, w_1, \ldots, w_l) = F(v_1, \ldots, v_k)G(w_1, \ldots, w_l). \tag{12.1}$$

$F \otimes G$ is an element of $L(V_1, \ldots, V_k, W_1, \ldots, W_l; \mathbb{R})$, called the ***tensor product of*** F ***and*** G.

Proposition 12.1.3 ▶ A Basis for the Space of Multilinear Functions

Let $V_1, \ldots, V_k$ be real vector spaces of dimensions $n_1, \ldots, n_k$, respectively. For each $j \in \{1, \ldots, k\}$, let $\left(E_1^{(j)}, \ldots, E_{n_j}^{(j)}\right)$ be a basis for V_j, and let $\left(\varepsilon_{(j)}^1, \ldots, \varepsilon_{(j)}^{n_j}\right)$ be the corresponding dual basis for V_j^*. Then the set

$$\mathcal{B} = \left\{\varepsilon_{(1)}^{i_1} \otimes \cdots \otimes \varepsilon_{(k)}^{i_k} : 1 \le i_1 \le n_1, \ldots, 1 \le i_k \le n_k\right\}$$

is a basis for $\mathrm{L}(V_1, \dots, V_k; \mathbb{R})$, which therefore has dimension equal to $n_1 \cdots n_k$.

Proof. • $\mathcal{B}$ ***spans*** $\mathrm{L}(V_1, \cdots, V_k; \mathbb{R})$: Suppose $F \in \mathrm{L}(V_1, \cdots, V_k; \mathbb{R})$ is arbitrary. Define $F_{i_1 \cdots i_k} = F\left(E^{(1)}_{i_1}, \cdots, E^{(k)}_{i_k}\right)$. Then for any k-tuples of vectors $(v_1, \cdots, v_k) \in V_1 \times \cdots \times V_k$, we write $v_1 = v_1^{i_1} E^{(1)}_{i_1}, \cdots, v_k = v_k^{i_k} E^{(k)}_{i_k}$, then

$$\begin{aligned} F(v_1, \cdots, v_k) &= v_1^{i_1} \cdots v_k^{i_k} F\left(E^{(1)}_{i_1}, \cdots, E^{(k)}_{i_k}\right) \\ &= v_1^{i_1} \cdots v_k^{i_k} F_{i_1 \cdots i_k} \\ &= F_{i_1 \cdots i_k} \varepsilon^{i_1}_{(1)} \otimes \cdots \otimes \varepsilon^{i_k}_{(k)} (v_1, \cdots, v_k) \end{aligned}$$

Thus

$$F = F_{i_1 \cdots i_k} \varepsilon^{i_1}_{(1)} \otimes \cdots \otimes \varepsilon^{i_k}_{(k)}$$

- $\mathcal{B}$ ***are linearly independent***: If the linear combinition of $\mathcal{B}$, say $F_{i_1 \cdots i_k} \varepsilon^{i_1}_{(1)} \otimes \cdots \otimes \varepsilon^{i_k}_{(k)}$ is equal to zero, by applying it on $\left(E^{(1)}_{j_1}, \cdots, E^{(k)}_{j_k}\right)$, we get $F_{j_1} \cdots F_{j_k} = 0$.

□

Covariant and Contravariant Tensors on a Vector Space

Definition 12.1.4

Let V be a finite-dimensional real vector space. If k is a positive integer, a ***covariant*** k***-tensor on*** V is an element of the k-fold tensor product $V^* \otimes \cdots \otimes V^*$, which we typically think of as a real-valued multilinear function of k elements of V:

$$\alpha : \underbrace{V \times \cdots \times V}_{k \text{ copies}} \to \mathbb{R}.$$

12.2 Symmetric and Alternating Tensors

Symmetric Tensors

Definition 12.2.1 ▸ Symmetric Tensors

Let V be a finite-dimensional vector space. A covariant k-tensor α on V is said to be ***symmetric*** if its value is unchanged by interchanging any pair of arguments:

$$\alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = \alpha(v_1, \dots, v_j, \dots, v_i, \dots, v_k)$$

whenever $1 \leq i < j \leq k$.

Definition 12.2.2 ▸ $\Sigma^k(V^*)$

The set of symmetric covariant k-tensors is a linear subspace of the space $T^k(V^*)$ of all covariant k-tensors on V; we denote this subspace by $\Sigma^k(V^*)$.

Definition 12.2.3 ▸ ${}^\sigma\alpha$

Given a k-tensor α and a permutation $\sigma \in S_k$, we define a new k-tensor ${}^\sigma\alpha$ by

$${}^\sigma\alpha(v_1, \dots, v_k) = \alpha\big(v_{\sigma(1)}, \dots, v_{\sigma(k)}\big).$$

Definition 12.2.4 ▸ Symmetrization

We define a projection

$$\operatorname{Sym} : \; T^k(V^*) \to \Sigma^k(V^*)$$

called ***symmetrization*** by

$$\operatorname{Sym} \alpha = \frac{1}{k!} \sum_{\sigma \in S_k} {}^\sigma\alpha.$$

More explicitly, this means that

$$(\operatorname{Sym} \alpha)(v_1, \dots, v_k) = \frac{1}{k!} \sum_{\sigma \in S_k} \alpha\big(v_{\sigma(1)}, \dots, v_{\sigma(k)}\big).$$

Definition 12.2.5 ▸ symmetric product

If $\alpha \in \Sigma^k(V^*)$ and $\beta \in \Sigma^l(V^*)$, we define their ***symmetric product*** to be the $(k+l)$-tensor $\alpha\beta$ (denoted by juxtaposition with no intervening product symbol) given by

$$\alpha\beta = \mathrm{Sym}(\alpha \otimes \beta).$$

More explicitly, the action of $\alpha\beta$ on vectors $v_1, \dots, v_{k+l}$ is given by

$$\alpha\beta(v_1, \dots, v_{k+l}) = \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} \alpha\big(v_{\sigma(1)}, \dots, v_{\sigma(k)}\big)\, \beta\big(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)}\big).$$

Alternating Tensors

Definition 12.2.6 ▸ k-covectors

We assume that V is a finite-dimensional real vector space. A covariant k-tensor α on V is said to be ***alternating*** (or ***antisymmetric*** or ***skew-symmetric***) if it changes sign whenever two of its arguments are interchanged. This means that for all vectors $v_1, \dots, v_k \in V$ and every pair of distinct indices i, j it satisfies

$$\alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = -\alpha(v_1, \dots, v_j, \dots, v_i, \dots, v_k).$$

Alternating covariant k-tensors are also variously called ***exterior forms***, ***multi-vectors***, or ***k-covectors***. The subspace of all alternating covariant k-tensors on V is denoted by $\Lambda^k(V^*) \subseteq T^k(V^*)$.

Definition 12.2.7

We define a projection $\mathrm{Alt} : T^k(V^*) \to \Lambda^k(V^*)$, called ***alternation***, as follows:

$$\mathrm{Alt}\,\alpha = \frac{1}{k!} \sum_{\sigma \in S_k} (\operatorname{sgn}\sigma)(\sigma\alpha),$$

where S_k is the symmetric group on k elements. More explicitly, this means

$$(\operatorname{Alt}\alpha)(v_1, \dots, v_k) = \frac{1}{k!} \sum_{\sigma \in S_k} (\operatorname{sgn}\sigma) \alpha\big(v_{\sigma(1)}, \dots, v_{\sigma(k)}\big).$$

12.3 The Algebra of Alternating Tensors

Definition 12.3.1

We assume on that V is a finite-dimensional real vector space. Given $\omega \in \Lambda^k(V^*)$ and $\eta \in \Lambda^l(V^*)$, we define their ***wedge product*** or ***exterior product*** to be the following $(k+l)$-covector:

$$\omega \wedge \eta = \frac{(k+l)!}{k!l!} \operatorname{Alt}(\omega \otimes \eta). \tag{12.2}$$

Remark.
系数的产生, 可以通过对 Alt 的另一种看法来理解:

- 先对所有 $(k+\ell)!$ 个排列求和.
- 但因为 α, β 已经交错, 同一个 (k,ℓ)-shuffle 等价类里的 $k!\,\ell!$ 个排列产出的值完全一样.
- 因此, Alt 本质上是

$$\frac{k!\,\ell!}{(k+\ell)!} \times \big(\text{对每个等价类选一个代表, 再对所有等价类求和}\big).$$

- 于是 wedge product 就等于

$$\alpha \wedge \beta = \frac{(k+\ell)!}{k!\,\ell!} \operatorname{Alt}(\alpha \otimes \beta) = \sum_{\text{等价类代表}} \operatorname{sgn}(\text{代表}) \cdot (\text{那一项}).$$

12.4 Tensors and Tensor Fields on Manifolds

12.5 Differential Forms on Manifolds

12.5.1 diffenrential k-form

Definition 12.5.1 ▸ diffenrential k-form

Let M be a n-dimensional smooth manifold(with or without boundary).

- The subset of T^kT^*M consisting of alternaint tensors is denoted by Λ^kT^*M:
$$\Lambda^kT^*M = \coprod_{p\in M} \Lambda^k\left(T_p^*M\right)$$
- A section of Λ^kT^*M is called a ***diffenrential k-form***, or just a ***k-form***. The integer k is called the ***degree*** of the form; this is a (continuous) tensor field whose value at each point is an alternating tensor.
- We denote the vector space of smooth k- forms by
$$\Omega^k(M) = \Gamma\left(\Lambda^kT^*M\right)$$
- If we define
$$\Omega^*(M) = \bigoplus_{k=0}^{n} \Omega^k(M)$$
then the wedge product turns $\Omega^*(M)$ into an associative, anditcommutative graded algebra.
- In any smooth chart, a k-form ω can be written locally as
$$\omega = \sum_I{}' \omega_I dx^{i_1}\wedge\cdots\wedge dx^{i_k} = \sum_I{}' \omega_I dX^I$$
where the coefficients ω_I are continuous functions defined on the coordinate domain, and we use dx^I as an abbreviation for $dx^{i_1}\wedge\cdots\wedge dx^{i_k}$

Definition 12.5.2 ▸ pullback of diffenrential forms

If $F : M \to N$ is a smooth map and ω is a differential form on N, then pullback $F^*\omega$ is a diffenrential form on M, defined as for any covariant tensor field:

$$(F^*\omega)_p (v_1, \cdots, v_k) = \omega_{F(p)} \left(dF_p (v_1), \cdots, dF_p (v_k)\right)$$

12.5.2 Exterior Derivatives

Definition 12.5.3

Let $U \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$, f is a 0-form on U, ω is a k-form on U. We define

- $$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i$$
- $$d\left(\sum_J{}' \omega_J dx^J\right) = \sum_J{}' d\omega_J \wedge dx^J$$

  In smoewhat more detail, this is

  $$d\left(\sum_J{}' \omega_J dx^{j_1} \wedge \cdots \wedge dx^{j_k}\right) = \sum_J{}' \sum_i \frac{\partial \omega_J}{\partial x^i} dx^i \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_k}$$

Proposition 12.5.4 ▸ Properties of the Exterior Derivative on $\mathbb{R}^n$

1. d is linear over $\mathbb{R}$.
2. If ω is a smooth k-form and η is a smooth l-form on an open subset $U \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$, then

   $$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta.$$

3. $d \circ d \equiv 0$.
4. d commutes with pullbacks: if U is an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, V is an open subset of $\mathbb{R}^m$ or $\mathbb{H}^m$, $F : U \to V$ is a smooth map, and $\omega \in \Omega^k(V)$, then

   $$F^*(d\omega) = d\left(F^*\omega\right). \tag{12.3}$$

Theorem 12.5.5 ▸ Existence and Uniqueness of Exterior Differentiation

Suppose M is a smooth manifold with or without boundary. There are unique operators $d : \Omega^k(M) \to \Omega^{k+1}(M)$ for all k, called ***exterior differentiation***, satisfying the following four properties:

(i) d is linear over $\mathbb{R}$.

(ii) If $\omega \in \Omega^k(M)$ and $\eta \in \Omega^l(M)$, then

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta.$$

(iii) $d \circ d \equiv 0$.

(iv) For $f \in \Omega^0(M) = C^\infty(M)$, df is the differential of f, given by $df(X) = Xf$.

In any smooth coordinate chart, d is given by Theorem 12.5.3.

Definition 12.5.6

- If $A = \bigoplus_k A^k$ is a graded algebra, a linear map $T : A \to A$ is said to be a ***map of degree m*** if $T(A^k) \subseteq A^{k+m}$ for each k.
- It is said to be an ***antiderivation*** if it satisfies $T(xy) = (Tx)y + (-1)^k x(Ty)$ whenever $x \in A^k$ and $y \in A^l$.

Remark.
Differential on functions extends uniquely to an antiderivation of $\Omega^* (M)$ of degree $+1$ whose square is zero.

12.5.3 An Invariant Formula for the Exterior Derivative

Proposition 12.5.7 ▸ Exterior Derivative of a 1-Form

For any smooth 1-form ω and smooth vector fields X and Y,

$$d\omega(X, Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y]). \tag{12.4}$$

Proposition 12.5.8 ▸ Invariant Formula for the Exterior Derivative

Let M be a smooth manifold with or without boundary, and $\omega \in \Omega^k(M)$.

For any smooth vector fields $X_1, \dots, X_{k+1}$ on M,

$$\begin{aligned}
&d\omega(X_1, \dots, X_{k+1}) \\
&= \sum_{1 \le i \le k+1} (-1)^{i-1} X_i \left(\omega(X_1, \dots, \hat{X}_i, \dots, X_{k+1})\right) \\
&\quad + \sum_{1 \le i < j \le k+1} (-1)^{i+j} \omega\left([X_i, X_j], X_1, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_{k+1}\right),
\end{aligned} \tag{12.5}$$

where the hats indicate omitted arguments.

12.5.4 Lie Derivatives of Differential Forms

Proposition 12.5.9

Suppose M is a smooth manifold, $V \in \mathfrak{X}(M)$, and $\omega, \eta \in \Omega^*(M)$. Then

$$\mathscr{L}_V(\omega \wedge \eta) = (\mathscr{L}_V \omega) \wedge \eta + \omega \wedge (\mathscr{L}_V \eta).$$

Theorem 12.5.10 ▸ Cartan's Magic Formula

On a smooth manifold M, for any smooth vector field V and any smooth differential form ω,

$$\mathscr{L}_V \omega = V \lrcorner (d\omega) + d(V \lrcorner \omega). \tag{12.6}$$

Corollary 12.5.11 ▸ The Lie Derivative Commutes with d

If V is a smooth vector field and ω is a smooth differential form, then

$$\mathscr{L}_V(d\omega) = d(\mathscr{L}_V \omega).$$

13 Orientations

13.1 Orientations of Manifolds

Definition 13.1.1

Let M be a smooth n-manifold with or without boundary, endowed with a pointwise orientations.

- If (E_i) is a local frame of TM, we say that (E_i) is ***(positively oriented)*** if $\left(E_1|_p, \cdots, E_n|_p\right)$ is positively oriented basis for T_pM at each point $p \in U$.A ***negatively oriented*** frame is defined analogously.
- A pointwise orientation is said to be ***continuous*** if every point of M is in the domain of an oriented local frame.
- An ***orientation of*** M is a continuous pointwise orientation.
- We say that M is ***orientable*** if there exists an orientation for it, and ***nonorientable*** if not.
- An ***oriented manifold*** is an ordered pair pra$M, \mathcal{O}$; an ***oriented manifold with boundary is defined similarly***.

14 Integration on Manifolds

14.1 Integration of Differential Forms

Proposition 14.1.1

Suppose D and E are open domains of integration in $\mathbb{R}^n$ or $\mathbb{H}^n$, and $G : \bar{D} \to \bar{E}$ is a smooth map that restricts to an orientation-preserving or orientation-reversing diffeomorphism from D to E. If ω is an n-form on $\bar{E}$, then

$$\int_D G^*\omega = \begin{cases} \displaystyle\int_E \omega & \text{if } G \text{ is orientation-preserving,} \\ \displaystyle -\int_E \omega & \text{if } G \text{ is orientation-reversing.} \end{cases}$$

14.2 Stoke's Theorem

Theorem 14.2.1 ▸ Stokes's Theorem

Let M be an oriented smooth n-manifold with boundary, and let ω be a compactly supported smooth $(n-1)$-form on M. Then

$$\int_M d\omega = \int_{\partial M} \omega. \tag{14.1}$$

15 De Rham Cohomology